MTH 405 Final term Notes

Lecture #23

Introduction to Metric Spaces.

* **Introduction and Definitions**
* Metric (meter, unit of distance)
* Space (set, in which some relation is defined)
* Metric space (x, distance function defined b/w its elements)

* **Distance:-**

x —— $d(x,y)$ —— y

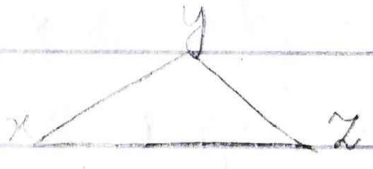
(i) $d(x,y) \geq 0$ (non-negativity)

(ii) $d(x,y) = 0 \iff x = y$ (Reflexive property)

(iii) $d(x,y) = d(y,x)$ (Symmetric property)

(iv) $d(x,y) + d(y,z) \geq d(x,z)$

(Triangular Inquality)



Question:-

Can there be a unified and Concise theory of distance??

Ans:- Yes, Metric Spaces be a unified and concise theory of distance.

Some Important Metrics and distance function:-

(i) usual Metric space $(\mathbb{R}, d)$

$x = \mathbb{R}$, $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$

$d(x, y) = |x, y|$

(ii) $H = \mathbb{R}^n$; $d = \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $x, y \in \mathbb{R}^n$

$$d(x, y) = \sum_{i=1}^{n} |x_i - y_i| \in \mathbb{R}$$

(iii) $x \neq \phi$ (Discret Metrics)

$$d_n(x, y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

(iv) $x = [a, b]$ = Class of all Continuous function on $[a, b]$

$d : X \times X \to \mathbb{R}$

$f, g \in X = [a, b]$

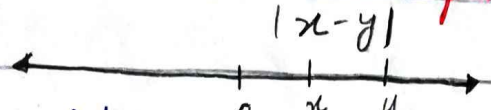
$$d(f, g) = \int_a^b |f(x) - g(x)| \, dx \quad \forall$$

Lecture # 24
Euclidean Metric

01- Usual Metric Space:-

let $x = \mathbb{R}$ & define $d: \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ as; $d(x,y) = |x-y|$, $x, y \in \mathbb{R}$.

Show that $(\mathbb{R}, d)$ is a metric space:-



(i) $d(x,y) = |x-y| \geq 0 \quad \forall\, x, y \in \mathbb{R}$

(ii) if $d(x,y) = 0$

$\Rightarrow |x-y| = 0$

Conversely; $x = y$

$x - y = 0$

$|x-y| = 0$

$d(x,y) = 0$

(iii) $d(x,y) = d(y,x)$

$d(x,y) = |x-y|$

$= |-(y-x)|$

$= |-1(y-x)| \quad \because |ab| = |a||b|$

$= +1|\,|y-x| \quad \because |-1| < 0 \Rightarrow -(-1) = 1$

$\Rightarrow |y-x| = d(y,x)$

$\therefore d(x,y) \leq d(x,z) + d(z,y)$

(iv) $d(x,y) = |x-y|$

$= |x - z + z - y|$

$\because \Rightarrow |a+b| \leq |a| + |b| \quad \forall a, b \in \mathbb{R}.$

$\therefore d(x,y) = |x - z + z - y|$

$\leq |x-z| + |z-y| = d(x,z) + d(z,y)$

$\therefore d(x,y) \leq d(x,z) + d(z,y)$

★ usual Metric in R_2

If $P_1(x_1, y_1), P_2(x_2, y_2) \in \mathbb{R}^2$, then show that :-

$$d(P_1, P_2) = +\sqrt{(x_1+x_2)^2 + (y_1+y_2)^2}$$

is a metric on $\mathbb{R}^2$.

★ Properties :-

M_1) $d(x,y) \geq 0$ (non-negativity)

M_2) $d(x,y) = 0 \Leftrightarrow x = y$ (Reflexive)

M_3) $d(x,y) = d(y,x)$ (Symmetry)

M_4) $d(x,y) \leq d(x,z) + d(z,y)$

Triangular inequality.

M1) :-

$(x_1 - x_2)^2 \geq 0$

$(y_1 - y_2)^2 \geq 0$

By adding :-

$(x_1 - x_2)^2 + (y_1 - y_2)^2 \geq 0$

$\sqrt{(x_1 - x_2)^2 + (y_1 + y_2)^2} \geq 0$

$d(P_1 P_2) \geq 0$

M2) :- $d(P_1, P_2) = 0$

$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = 0$

let :-

$\Leftrightarrow (x_1 - x_2)^2 + (y_1 - y_2)^2 = 0$

$\Rightarrow (x_1 - x_2)^2 = 0 \quad \& \quad (y_1 - y_2)^2 = 0$

$\Leftrightarrow x_1 - x_2 = 0 \quad , \quad y_1 - y_2 = 0$

$\Leftrightarrow x_1 = x_2 \quad , \quad y_1 = y_2$

$\Leftrightarrow (x_1, y_1) = (x_2, y_2)$

$\Rightarrow P_1 = P_2$

M3) :- $d(P_1, P_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = d(P_2, P_1)$

$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $\{(x-y)^2 = (y-x)^2$

$= d(P_2, P_1)$

M4) :- $d(P_1, P_2) \leq d(P_1, P_3) + d(P_3, P_2)$

All Points are on the same line.

Case 1:-

if P_1, P_2, P_3 are **collinear.**

$$d(P_1, P_2) = d(P_1, P_3) + d(P_3, P_2) \rightarrow (1)$$

Case 2:-

if P_1, P_2, P_3 are non-collinear

$$d(P_1, P_2) < d(P_1, P_3) + d(P_3, P_2) \rightarrow (2)$$

on Combining (1) and (2)

$$d(P_1, P_2) \leq d(P_1, P_3) + d(P_3, P_2)$$

$\therefore (\mathbb{R}^2, d)$ is a metric Space.

Lecture #25
Taxicab and Discrete Metric Spaces.

Discrete Metric Space:-

Show that any non-empty set X, can be made a metric space (x, d_0), where $d : X \times X \to \mathbb{R}$ is defined by:-

$$d_0(x,y) = \begin{cases} 0 & \text{if } x=y \\ 1 & \text{if } x \neq y \end{cases}$$

m_1):- $d_0(x,y) = 0$ if $x = y$
$d_0(x,y) = 1$ if $x \neq y$ } By def

On Combining both:-

$$d_0(x,y) \geq 0$$

m_2):- if $(x,y) = 0 \Rightarrow x = y$
if $x = y \Rightarrow d_0(x,y)$ } by def

m_3):- Case i:-

if $x = y \Rightarrow d_0(x,y) = 0 = d_0(y,x)$

Case ii:-

if $x \neq y \Rightarrow d_0(x,y) = 1 = d_0(x,y)$

$\therefore d_0(x,y) = d_0(y,x)$

$d_0(x,y) \leq d_0(x,z) + d_0(z,y)$

Proof:-

$x, y, z \in X \neq \phi$

$x = y = z$, $x \neq y = z$, $x \neq y \neq z$.

Case i;

if $x = y = z \Rightarrow d(x,y) = d(y,z)$

$= d(z,x) = 0$

$d(x,y) \leq d_0(x,z) + d_0(z,y) \rightarrow (4)$

$0 = 0 + 0$

Case ii:- if $x = y \neq z \Rightarrow x \neq z$

$d_0(x,y) = 0$; $d_0(y,z) = 1$.

$d_0(x,z) = 1$

$\therefore (1) \Rightarrow 0 < 1 + 1 \rightarrow (2)$

Case (iii):-

if $x \neq y \neq z$

$d(x,y) = d(y,z) = d(z,y) = 1$

$\therefore (1) \Rightarrow 1 < 1 + 1$

Combining all three cases;

$d_0(x,y) \leq d_0(x,z) + d_0(z,y)$

Taxicab - 1 :-

Show that $d(P_1, P_2) = |x_1 - x_2| + |y_1 - y_2|$ is a metric on $\mathbb{R}^2$, where $P_1(x_1, y_1)$, $P_2(x_2, y_2) \in \mathbb{R}^2$.

m_1) :-

$|x_1 - x_2| \geq 0 \quad \forall x_1, x_2 \in \mathbb{R}$

$|y_1 - y_2| \geq 0 \quad \forall y_1, y_2 \in \mathbb{R}$

$|x_1 - x_2| + |y_1 - y_2| \geq 0$

$\Rightarrow d(P_1, P_2) \geq 0$

m_2) :-

$d(P_1, P_2) = 0$

$\Leftrightarrow |x_1 - x_2| + |y_1 - y_2| = 0$

$\Leftrightarrow |x_1 - x_2| = 0 \quad \& \quad |y_1 - y_2| = 0$

$\Leftrightarrow x_1 - x_2 = 0 \quad \& \quad y_1 - y_2 = 0$

$x_1 = x_2 \quad , \quad y_1 = y_2$

$\Leftrightarrow (x_1, y_1) = (x_2, y_2)$

$P_1 = P_2$

Taxicab - 2 :-

m_3) :-

$d(P_1, P_2) = |x_1 - x_2| + |y_1 - y_2|$

$= |-(x_2 - x_1)| + |-(y_2 - y_1)|$

$= |x_2 - x_1| + |y_2 - y_1|$

$= d(P_2, P_1)$

My :- $d(P_1, P_2) = |x_1 - x_2| + |y_1 - y_2|$

$= |x_1 - x_3 + x_3 - x_2| + |y_1 - y_3 + y_3 - y_2|$

$\Rightarrow |a+b| \leq |a| + |b|$

$\leq |x_1 - x_3| + |x_3 - x_2| + |y_1 - y_3| + |y_3 - y_2|$

$= (|x_1 - x_3| + |y_1 - y_3|) + [|x_3 - x_2| + |y_3 - y_2|]$

$= d(P_1, P_3) + d(P_3, P_2)$

$d(P_1, P_2) \leq d(P_1, P_3) + d(P_3, P_2)$

★ Taxicabe In R_n-1:-

Let $x = (x_1, x_2, \cdots, x_n)$, $y = (y_1, y_2, \cdots y_n) \in \mathbb{R}^n$ then Show that $d(x,y) = \sum_{i=1}^{n} |x_i - y_i|$ is a metric on $\mathbb{R}^n$.

My :- $d(x, y) \geq 0$

$d(x, y) = \sum_{i=1}^{n} |x_i - y_i|$

$= |x_1 - y_1| + |x_2 - y_2| + \cdots + |x_n - y_n| \geq 0$

$d(x, y) \geq 0$

My :-

$d(x, y) = 0$

$\Rightarrow \sum_{i=1}^{n} |x_i - y_i| = 0$

$|x_1 - y_1| + |x_2 - y_2| + \cdots + |x_n - y_n| = 0$

$|x_1 - y_1| = 0 \;, |x_2 - y_2| = 0 \cdots, |x_n - y_n| = 0$

$x_1 - y_1 = 0 \;, x_2 - y_2 = 0 \cdots x_n - y_n = 0$

$x_1 = y_1, \quad x_2 = y_2, \quad \cdots \quad x_n = y_n$

$\therefore (x_1, x_2, \cdots, x_n) = (y_1, y_2, \cdots, y_n)$

$\therefore x = y$

My :- $d(x,y) = d(y,x)$

$$d(x,y) = \sum_{i=1}^{n} |x_i - y_i|$$

$$= \sum_{i=1}^{n} |y_i - x_i|$$

$$= d(y,x)$$

My :- $d(x,y) \leq d(x,z) + d(z,y)$

$$d(x,y) = \sum_{i=1}^{n} |x_i - y_i|$$

$$= \sum_{i=1}^{n} |x_i - z_i + z_i - y_i|$$

$$= |(x_1 - z_1) + (z_1 - y_1)| + |(x_2 - z_2) + (z_2 - y_2)| + \cdots + |(x_n - z_n) + (z_n - y_n)|$$

$\because = |a_1 + a_2| \leq |a_1| + |a_2|, \forall a_1, a_2 \in \mathbb{R}.$

$$\leq |x_1 - z_1| + |z_1 - y_1| + |x_2 - z_2| + |z_2 + y_2| + \cdots + |x_n - z_n| + |z_n + y_n|$$

$$= \{|x_1 - z_1| + |x_2 - z_2| + \cdots + |x_n - z_n|\} + \{|z_1 + y_1| + |z_2 + y_2| + \cdots + |z_n - y_n|\}$$

$$= \sum_{i=1}^{n} |x_i - z_i| + \sum_{i=1}^{n} |z_i - y_i|$$

$$= d(x,z) + d(z,y)$$

$d(x,y) \leq (x,z) + d(z,y)$

Lecture # 26:- Metric in terms of defined Metric

- Metric on $\mathbb{R} - \{0\}$

Show that $d: \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defined by $d(x,y) = \left|\frac{1}{x} - \frac{1}{y}\right|$, $\forall x, y \in \mathbb{R} - \{0\}$ is a metric on $\mathbb{R} - \{0\}$,

m_1:- $\left|\frac{1}{x} - \frac{1}{y}\right| \geq 0 \quad \forall x, y \in \mathbb{R} - \{0\}$.

$\Rightarrow d(x,y) \geq 0$

m_2:- $d(x,y) = 0$

$\Leftrightarrow \left|\frac{1}{x} - \frac{1}{y}\right| = 0$

$\Leftrightarrow \frac{1}{x} - \frac{1}{y} = 0$

$\Leftrightarrow \frac{1}{x} = \frac{1}{y} \Leftrightarrow x = y \quad \forall x, y \in \mathbb{R} - \{0\}$

m_3:- $d(x,y) = \left|\frac{1}{x} - \frac{1}{y}\right|$

$= \left|-\left(\frac{1}{y} - \frac{1}{x}\right)\right|$

$= \left|\frac{1}{y} - \frac{1}{x}\right|$

$d(x,y) = d(y,x)$

$d(x,y) = \frac{1}{x-a} - \frac{1}{y-a}$ $\mathbb{R} - \{a\}$

My :- $d(x,y) \leq d(x,z) + d(z,y)$

$d(x,y) = \left|\frac{1}{x} - \frac{1}{y}\right|$

$= \left|\frac{1}{x} - \frac{1}{z} + \frac{1}{z} - \frac{1}{y}\right|$

$d(x,y) \leq \left|\frac{1}{x} - \frac{1}{z}\right| + \left|\frac{1}{z} - \frac{1}{y}\right|$

$d(x,y) \leq d(x,z) + d(z,y)$

British Rail distance function :-

Let $d'(x,y) = |x| + |y|$; $x, y \in \mathbb{R}$, Show that d' is not a metric on $\mathbb{R}$.

My :-

$d'(x,y) = |x| + |y|$

$\because |x| \geq 0 \; ; |y| \geq 0 \Rightarrow d'(x,y) \geq 0$

Ma :-

$d'(x,y) = 0$

$\Rightarrow |x| + |y| = 0$

$\Rightarrow |x| = 0 \; ; |y| = 0$

$x = 0 \; ; \; y = 0$

$x = 0 = y \Rightarrow \boxed{x = y}$

Conversly :-

$\because x = y$

$\not\Rightarrow x = y = 0$

$\therefore d'(x,y)$ is not a metric.

★ Absolute and radical Metric:-

Show that the function $d: \mathbb{R}\times\mathbb{R}\to\mathbb{R}$ defined by $d(x,y) = \sqrt{|x-y|}$, $\forall x,y\in\mathbb{R}$ is a metric on $\mathbb{R}$

M_1 :- $\because |x-y| \geq 0$

$\Rightarrow \sqrt{|x-y|} \geq 0$

$\Rightarrow d(x-y) \geq 0$

M_2 :- $d(x,y) = 0$

$\Leftrightarrow \sqrt{|x-y|} = 0$

$\Leftrightarrow |x-y| = 0$

$\Leftrightarrow x-y = 0$

$\Leftrightarrow x = y$

M_3 :- $d(x,y) = d(y,x)$

$= \sqrt{|x-y|} = \sqrt{|-(y-x)|}$

$= \sqrt{|y-x|} = d(y,x)$

M_4 :-

$d(x,y) = \sqrt{|x-y|}$

$\Rightarrow |a+b| \leq |a| + |b|$

$|a+b| \leq |a| + |b| + 2\sqrt{a}\sqrt{b}$

$|a+b| < \left(\sqrt{|a|} + \sqrt{|b|}\right)^2$

$\sqrt{|a+b|} \leq \sqrt{|a|} + \sqrt{|b|}$ → (1)

Put :-

$$a = x - z, \quad y = z - y$$

(1) $\Rightarrow \sqrt{|x - z + z - y|} \leq \sqrt{|x - z|} + \sqrt{|z - y|}$

$\Rightarrow \sqrt{|x - y|} \leq \sqrt{|x - z|} + \sqrt{|z - y|}$

$\Rightarrow d(x, y) = \sqrt{|x - y|}$ is a Metric on $\mathbb{R}$.

* Metric Interms of defined metrics:-

Let (x, d) be metric space, Show that d' defined by:-

$$d'(x, y) = \frac{d(x, y)}{1 + d(x, y)}$$ **is also metric on X.**

My:- $\because$ 'd' is metric on 'X'

$\Rightarrow d(x, y) \geq 0 \quad \forall\, x, y \in X$

$\Rightarrow 1 + d(x, y) \geq 0$

$\Rightarrow \frac{d(x, y)}{1 + d(x, y)} \geq 0 \Rightarrow d'(x, y) \geq 0;$

$\forall\, x, y \in \mathbb{R}$

M_2 :- $d'(x,y) = 0$

$$\frac{d(x,y)}{1+d(x,y)} = 0 \Rightarrow d(x,y) = 0$$

$\because$ 'd' is Metric on X

$\Rightarrow d(x,y) = 0 \Rightarrow x = y$

Conversely: if $x = y \Rightarrow d(x,y) = 0$

$$\Rightarrow \frac{d(x,y)}{1+d(x,y)} = 0 \Rightarrow d'(x,y) = 0$$

M_3 :-

$$d'(x,y) = \frac{d(x,y)}{1+d(x,y)}$$

$\because$ d is Metric on 'X'

$\Rightarrow d(x,y) = d(y,x)\ \forall$

$$\Rightarrow d'(x,y) = \frac{d(x,y)}{1+d(y,x)} = d'(y,x)$$

M_4 :-

$$d'(x,y) \leq d'(x,z) + d'(z,y)$$

$\because$ 'd' is Metric on X.

$\Rightarrow d(x,y) \leq d(x,z) + d(z,y), \forall x,y,z \in X$

$$\Rightarrow 1 + d(x,y) \leq 1 + d(x,z) + d(z,d)$$

$$\frac{1}{1+d(x,z)+d(z,y)} \leq \frac{1}{1+d(x,y)}$$

$$1 - \frac{1}{1+d(x,y)} \leq 1 - \frac{1}{1+d(x,z)+d(z,y)}$$

$$\frac{1+d(x,y)-1}{1+d(x,y)} \leq \frac{1+d(x,z)+d(z,y)-1}{1+d(x,z)+d(z,y)}$$

$$\frac{d(x,y)}{1+d(x,y)} \leq \frac{d(x,z)+d(z,y)}{1+d(x,z)+d(z,y)}$$

$$d'(x,y) \leq \frac{d(x,z)}{1+d(x,z)+d(z,y)} + \frac{d(z,y)}{1+d(x,z)+d(z,y)}$$

X Repeated

$$\Rightarrow \frac{1+d(x,y)-1}{1+d(x,y)} \leq \frac{1+d(x,z)+d(z,y)-1}{1+d(x,z)+d(z,y)}$$

$$\Rightarrow \frac{d(x,y)}{1+d(x,y)} \leq \frac{d(x,z)+d(z,y)}{1+d(x,z)+d(z,y)}$$

$$\Rightarrow d'(x,y) \leq \frac{d(x,z)}{1+d(x,z)} + \frac{d(z,y)}{1+(z,y)}$$

$$d'(x,y) \leq d'(x,z) + d'(z,y)$$

Lecture # 27

Chebyshev Metric and Metric on Continuous Functions.

* Chebyshev Metric (in R^2-1)

Show that $d(P_1, P_2) = \max\{|x_1 - x_2|, |y_1 - y_2|\}$

$\forall\ P_1(x_1, y_1);\ P_2(x_2, y_2) \in R^2$.

$d: R^2 \times R^2 \to R$ is a Metric Space (d, R^2)

$d(P_1, P_2) = \max\{|x_1 - x_2|, |y_1 - y_2|\}$

My:- $d(x, y) \geq 0$

$|x_1 - x_2| \geq 0$ (I)

$|y_1 - y_2| \geq 0$ (II)

from (I) and (II)

$\max\{|(x_1 - x_2)|, |y_1 - y_2|\} \geq 0$

$d(P_1, P_2) \geq 0$

My:- $d(x_1, y_1) = 0$

$|x_1 - x_2| = 0$, $|y_1 - y_2| = 0$

$x_1 - x_2 = 0$ $\qquad$ $y_1 - y_2 = 0$

$x_1 = x_2$ $\qquad$ $y_1 = y_2$

$(x_1, y_1) = (x_2, y_2)$

$P_1 = P_2$

$d(P_1, P_2) = 0$

m3) :-

$$d(x,y) = d(y,x)$$

$$d(P_1,P_2) = \max\{|x_1-x_2|, |y_1-y_2|\}$$

$$= \max\{|-(x_2-x_1)|, |-(y_2-y_1)|\}$$

$$= \max\{|x_2-x_1|, |y_2-y_1|\}$$

$$= d(P_2,P_1)$$

m4) :-

$$d(P_1,P_2) \leq d(P_1,P_3)+d(P_3,P_1)$$

$$d(P,P_2) = \max\{|x_1-x_2|, |y_1-y_2|\}, P_3(x_3,y_3) \in \mathbb{R}^2$$

$$= \max\{|x_1-x_3+x_3-x_2|, |y_1-y_3+y_3-y_2|\}$$

$$= \max\{|(x_1-x_3)+(x_3-x_2)| + |(y_1-y_3)+(y_3-y_2)|\}$$

$$|a+b| \leq |a|+|b|$$

$$|(x_1-x_3)+(x_3-x_2)| \leq |x_1-x_3|+|x_3-x_2|$$

$$|(y_1-y_3)+(y_3-y_2)| \leq |y_1-y_3|+|y_3-y_2|$$

$$d(P_1,P_2) \leq \max\{|x_1-x_3|+|x_3-x_2|+|y_1-y_3|+ |y_3-y_2|\} \rightarrow (2)$$

If $a,b,c,d \in \mathbb{R}$

$$a \leq \max\{a,b\}, \quad c \leq \max\{c,d\} \rightarrow (A)$$

$$b \leq \max\{a,b\}, \quad d \leq \max\{c,d\} \rightarrow (B)$$

By adding (A)

$$a+c \leq \max\{a,b\}+\max\{c,d\}$$

By adding (B)

$$b+d \leq \max\{a,b\}+\max\{c,d\}$$

$\Rightarrow \max\{a+c, b+d\} \leq \max\{a,b\} + \max\{c,d\}$

$\therefore$ (2) $\Rightarrow$

$d(P_1,P_2) \leq \max\{|x_1-x_3|, |y_1-y_3|\} + \max\{|x_3-x_2|, |y_3-y_2|\}$

$\Rightarrow d(P_1,P_2) \leq d(P_1,P_3) + d(P_3,P_2)$

$\therefore (d\mathbb{R}^2)$ is a metric space.

Chebyshev Metric in $\mathbb{R}^n$:-

Let $X = \mathbb{R}^n$, $P_1 = (x_1, x_2, \cdots x_n)$

$P_2 = (y_1, y_2, \cdots, y_n) \in \mathbb{R}^n$ then prove that $d: \mathbb{R}^n \times \mathbb{R}^n \to R$ is a metric on $\mathbb{R}^n$ defined by $d(P_1,P_2)$

$= \max\{|x_i - y_i|; 1 \leq i \leq n, n \in \mathbb{Z}^+\}$.

$P_1 = (x_1, x_2 + \cdots, x_n)$, $P_2 = (y_1, y_2 + \cdots, y_n) \in \mathbb{R}^n$

given that $d(P_1,P_2) = \max\{|x_i - y_i|, 1 \leq i \leq n, n \in \mathbb{Z}^+\}$

$\Rightarrow d(P_1,P_2) = \max\{|x_1 - x_2|, |x_2 - y_2|, \cdots, |x_n - y_1|\}$ $\to$ (1)

M1:- $\because |x_1-x_2| \geq 0,\ |x_2-y_2| \geq 0, \cdots, |x_n-y_n|$

$\max\{|x_1-x_2|, |x_2-y_2|, \cdots, |x_n-y_n|\} \geq 0$

$\max\{|x_i-y_i|, 1 \leq i \leq n\} \geq 0$

$\Rightarrow \quad d(P_1, P_2) \geq 0$

M2:-

$d(P_1, P_2) = 0$

$\Rightarrow \max\{|x_i-y_i|, 1 \leq i \leq n\} = 0$

$\Rightarrow \max\{|x_1-y_1|, |x_2-y_2|, \cdots, |x_n-y_n|\} = 0$

$\Rightarrow |x_1-y_1| = 0,\ |x_2-y_2| = 0, \cdots |x_n-y_n| = 0$

$\Rightarrow x_1-y_1 = 0,\ x_2-y_2 = 0 \cdots x_n-y_n = 0$

$\Rightarrow x_1 = y_1 \quad x_2 = y_2 \cdots x_n = y_n$

$(x_1, x_2, \cdots x_n) = (y_1, y_2, \cdots y_n)$

$\Rightarrow \quad P_1 = P_2$

M3:-

$d(P_1, P_2) = \max\{|x_i-y_i|;\ 1 \leq i \leq n\}$

$= \max\{x_1-y_1|, |x_2-y_2|, \cdots, |x_n-y_n|\}$

$= \max\{|-(y_1-x_1)|, |-(y_2-x_2)|, \cdots, |-(y_n-x_n)|\}$

$\because |-a| = |a| \quad \forall\ a \in \mathbb{R}.$

$= \max\{|y_1-x_1|, |y_2-x_2|, \cdots |y_n-x_n|\}$

$= \max\{|y_i-x_i|;\ 1 \leq i \leq n\}$

$= d(P_2, P_1)$

My:- Now we will prove triangular Inequality for $P=(x_1, x_2 \cdots x_n)$, $q=(y_1, y_2 \cdots y_n)$, $r=(z_1, z_2 \cdots z_n) \in \mathbb{R}^n$.

$d(P,q) \le d(P,T)+d(r,q)$

$d(P,q) = \max\{|x_i-y_i|; 1 \le i \le n, n \in \mathbb{Z}^+\}$

$= \max\{|x_1-y_1|, |x_2-y_2| \cdots, |x_n-y_n|\}$

$= \max\{|x_1-z_n+z_n-y_1|, |x_2-z_2+z_2-y_2| \cdots |x_n-z_n+z_n-y_n|\} \to (1)$

$|x_1-z_1+z_1-y_1| \le |x_1-z_1|+|z_1-y_1|$

$|x_2-z_2+z_2-y_2| \le |x_2-z_2|+|z_2-y_2|$

$\vdots$

$|x_n-z_n+z_n-y_n| \le |x_n-z_n|+|z_n-y_n|$

$\Rightarrow \max\{|x_1-z_1+z_1-y_1|, |x_2-z_2+z_2-y_2|, \cdots, |x_n-z_n+z_n-y_n|\} \le \max\{|x_1-z_1|+|z_1-y_1|, |x_2-z_2|+|z_2-y_2| \cdots |x_n-z_n|+|z_n-y_n|\} \to (2)$

If $a_1, a_2 \cdots a_n \in \mathbb{R}$ & $b_1, b_2 \cdots b_n \in \mathbb{R}$

$\Rightarrow a_1 \le \max\{a_1, a_2 \cdots a_n\}$, $b_1 \le \max\{b_1, b_2 \cdots b_n\}$

$\Rightarrow a_2 \le \max\{a_1, a_2 \cdots a_n\}$, $b_2 \le \max\{b_1, b_2 \cdots b_n\}$

$\Rightarrow a_n \le \max\{a_1, a_2, \cdots, a_n\}$, $b_n \le \max\{b_1, b_2 \cdots b_n\}$

$a_1+b_1 \le \{a_1, a_2 \cdots a_n\} + \max\{b_1, b_2 \cdots b_n\}$

$a_2+b_2 \le \max\{a_1, a_2, \cdots a_n\} \max\{b_1, b_2 \cdots b_n\}$

$\vdots$

$a_n+b_n \le \max\{a_1, a_2 \cdots a_n\} + \max\{b_1, b_2 \cdots b_n\}$

$\max\{a_1+b_1, a_2+b_2, \dots a_n+b_n\} \le \max\{a_1, a_2, \dots a_n\}$
$+ \max\{b_1, b_2, \dots b_n\}$

$\Rightarrow$

$d(P,Q) \le \max\{|x_1-z_1|, |x_2-z_2|, \dots, |x_n-z_n|\} + \max\{|z_1-y_1|, |z_2-y_2| \dots |z_n-y_n|\}$

$\le \max\{|x_i-z_i|, 1 \le i \le n\} + \max\{|z_i-y_i| : 1 \le i \le n\}$

$d(P,Q) \le d(P,R) + d(R,Q)$

$\therefore (d, \mathbb{R}^n)$ is a metric space.

C[a,b] as a Metric space -1:-

for C[a,b], the set of all real Continuous functions defined on [a,b] with $d(f,g) = \int_a^b |f(x)-g(x)|dx$, $(b>a)$ for all $f(x), g(x) \in C[a,b]$.

Show that $(C[a,b], d)$ is a Metric space:-

m_1:-

$$d(f,g) = \int_a^b |f(x) - g(x)|\,dx$$

$\forall\ x \in [a,b]$

$\because |f(x) - g(x)| \geq 0$

$$\Rightarrow \int_a^b |f(x) - g(x)|\,dx \geq 0$$

$$\Rightarrow d(f,g) \geq 0$$

m_2:- $d(f,g) = 0$

$$\Rightarrow \int_a^b |f(x) - g(x)|\,dx = 0$$

$$\Rightarrow |f(x) - g(x)| = 0$$

$$\Rightarrow f(x) = g(x)$$

$$\Rightarrow f = g$$

m_3:-

$$d(f,g) = \int_a^b |f(x) - g(x)|\,dx$$

$$= \int_a^b |-\{g(x) - f(x)\}|\,dx$$

$$= \int_a^b |g(x) - f(x)|\,dx = d(g,f)$$

m_4:-

$$d(f,g) = \int_a^b |f(x) - g(x)|\,dx$$

$$= \int_a^b |f(x) - h(x) + h(x) - g(x)|\,dx \rightarrow (1)$$

$$= \because f(x) = |f(x) - h(x) + h(x) - g(x)| \leq |f(x) - h(x)| + |h(x) - g(x)|$$

$$\Rightarrow \int_a^b |f(x) - h(x) + h(x) - g(x)|\,dx \le \int_a^b |f(x) - h(x)| + |h(x) - g(x)|\,dx$$

$$= \int_a^b |f(x) - h(x)|\,dx + \int_a^b |h(x) - g(x)|\,dx$$

$$d(f,g) \le d(f,h) + d(h,g)$$

Example:-

Determine whether the given distance functions are the metrices on $\mathbb{R}$.

i) $d(x,y) = |x^2 - y^2|$

m_1 :- Non-negativity :-

$\because |x^2 - y^2| \ge 0 \Rightarrow d(x,y) \ge 0$

m_2 :- Reflexive property :-

if $d(x,y) = 0 \Rightarrow |x^2 - y^2| = 0$

$x^2 - y^2 = 0, \quad x^2 = y^2$

$\Rightarrow x = \pm y \Rightarrow d(x,y) = 0 \Rightarrow x = y$

but also $\Rightarrow$ $x = -y$

$\Rightarrow$ Reflexive property is not satisfied

$\Rightarrow$ given ~~property~~ d is not a Metric on $\mathbb{R}$.

(ii) $d(x,y) = |x-y|^2$.

m_1 :- Non-Negtivity:-

$|x-y|^2 \geq 0 \Rightarrow d(x,y) \geq 0$

m_2 :- Replexive Property:-

if $d(x,y) = 0$

$\Rightarrow |x-y|^2 = 0$, $|x-y| = 0 \Rightarrow x-y=0$

$\Rightarrow x = y$

m_3) Symmetric Property:-

$d(x,y) = |x-y|^2 = |y-x|^2 = d(y,x)$

m_4 Triangular Inequality:-

if $x=1$, $y=2$, $z=3$

$d(1,3) = |1-3|^2 = 4$.

$d(1,2) = |1-2|^2 = 1 = d(2,3)$

$\therefore d(1,3) > d(1,2) + d(2,3)$

$\Rightarrow$ Triangular Inequality is not satisfied $\therefore (d, \mathbb{R})$ is not a Metric Space.

Mth405- Lec #28

Uniform metric space on class of continuous funct

Distance formula

If $B[a,b]$, $a<b$ is the set of all real valued bounded function defined on $[a,b]$ Then show that distance function.

$d(f,g) = \sup|f(x)-g(x)| \quad \forall\ f,g \in B[a,b]$

Bounded function:

which is in/between close interval

$f(n)=(-1)^n$, $F(x)=(-1)^x$, $n \in \mathbb{Z}^+$

which show $[-]$.

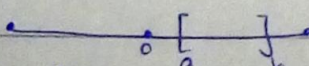
Unbounded function:

Which lie in open-interval $= \left[\begin{matrix} x \to -\infty \\ e^x \to 0 \end{matrix}\right]$ $0 \le e^x < \infty$

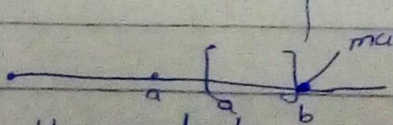
Mathematically

Superimum & Like maximum.

e.g



What is the maximum value of closed interval $[a,b]$.



Above the real line value on the right side is higher on the left

Case II : $d'(x,y) = d(x,y)$ $\quad d(x,y) < 1$ 'd is metric

$d(y,x) \quad d(y,x) < 1$

$d'(x,y) = \min\{1, d(y,x) = d'(y,x)$ — B

From A, B $d'(x,y) = d'(y,x) \quad \forall x,y \in X \neq \phi$

iii) Triangular $d(x,y) \leq d(x,z) + d(x,y) \quad \forall x,y,z \neq \phi$

Case I $d'(x,y) = \min\{1, d(x,y)\}$

$= d(x,y)$

$= d(x,z) + d(z,y)$

$= \min\{1, d(x,z) + d(z,y)$ — A

Case II $\min\{1, d(xz)\} = I$, $\min\{1, d(z,y)\} = 1$ — II

$\min\{1, d(x,z)\} = d'(x,z)$ — III

$\min\{1, d(z,y)\} = d'(z,y)$ — IV

Add I, II $\min\{1, d(x,z)\} + \min\{1, d(z,y)\} = 1+1 = 2$ — (a)

Add I, Iv $\min\{1, d(x,z)\} + \min\{1, d(z,y)\} = 1 + d'(z,y)$ — (b)

Add II, III $\min\{1, d(x,z)\} + \min\{1, d(z,y)\} = d'(x,z) + 1$ — (c)

Add III, IV $\min\{1, d(x,z)\} + \min\{1, d(z,y)\} = d'(x,z) + d'(z,y)$ — (d)

From a, b, c, d

$\min\{1, d(x,z)\} + \min\{1, d(z,y)\} = \{a, b, c, d\}$

$\min\{1, d(x,z)\} + \min\{1, d(z,y)\} = \{1+1, 1+d'(z,y)\} =$

$\{1+1, 1+d(z,y), d'(x,z)+1, d'(x,z) + d(z,y)\}$

From A, B.

$\min\{1, d(x,z) + d(z,y)\} \leq \min\{1, d(x,z)\} + \min\{1, d(z,y)\}$

If $p=(x_1,x_2), q=(y_1,y_2) \in \mathbb{R}^2$ Show that.

$d(p,q) = \min\{|x_1-y_1|, |x_2-y_2|\}$ is metric on $\mathbb{R}^2$

Proof m_1) non-negative $d(x,y) \geq 0 \ \forall x,y \in X$

$|x_1-y_1| \geq 0 - I$, $|x_2-y_2| \geq 0 - II$

From I, II $\min\{|x_1-y_1|, |x_2-y_2|\} \geq 0$,

$d(p,q) \geq 0 \quad \forall x,y \in \mathbb{R}^2$

m_2) Reflexive $d(x,y)=0 \quad x=y$

$d(p,q)=0$, $\min\{|x_1-y_1|, |x_2-y_2|\}=0$

case I $|x_1-y_1|=0$ Then $|x_2-y_2|>0$

$x_1-y_1=0$ $x_2-y_2>0$

$x_1=y_1$ $x_2>y_2$

$x_2 \neq y_2$

$p \neq q$ $x_2 \neq y_2$ not true for reflexive.

Hence

$d(p,q) = \min\{|x_1-y_1|, |x_2-y_2|\}$ not metric

*=

If $p=(x_1,x_2), q=(y_1,y_2) \in \mathbb{R}^2$. Show that $d(p,q) = |x_1|+|x_2|+|y_1|+|y_2|$ is metric space on $\mathbb{R}^2$.

Proof m_1) - non-negative $d(x,y) \geq 0 \ \forall x,y \in$

$|x_1| \geq 0, |x_2| \geq 0, |y_1| \geq 0, |y_2| \geq 0.$

$|x_1|+|x_2|+|y_1|+|y_2| \geq 0$

$d(p,q) \geq 0$

2) Reflexive $d(x,y)=0 \quad x=y \quad x,y \in X \neq \phi$

$d(p,q)=0, \; |x_1|+|x_2|+|y_1|+|y_2|=0,$

$|x_1|=0, \; |x_2|=0, \; |y_1|=0, \; |y_2|=0$

$x_1=0, \; x_2=0, \; y_1=0, \; y_2=0,$

$x_1=x_2, \quad y_1=y_2$

$(x_1,y_1)=(x_2,y_2)$

Converse

$|x_1| \neq 0, \; |x_2| \neq 0, \; |y_1| \neq 0, \; |y_2| \neq 0$

Mth 405

lec 30

Theorem

If $(x_1, x_2, \cdots x_n), (y_1, y_2, \cdots y_n) \in R^n$

Cauchy Schwarz

$$\left(\sum_{r=1}^{n} x_r y_r\right)^2 \leq \left(\sum_{r=1}^{n} x_r^2\right) \cdot \left(\sum_{r=1}^{n} y_r^2\right)$$

Let proof

$\vec{u} = (x_1, x_2, \cdots x_n)$, $\vec{v} = (y_1, y_2, \cdots y_n) \in R^n$

Apply data product

$\vec{u} \cdot \vec{v} = |u||v| \cos\phi$ (ϕ is equal to b/w two ve

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \sqrt{x_1^2 + x_2^2 + x_n^2} \cdot \sqrt{y_1^2 + y_2^2 + \cdots y_n^2} \cos\theta$$

$$x_1 y_1 + x_2 y_2 + \cdots x_n y_n = \sqrt{(x_1^2 + x_2^2 + \cdots x_n^2) \cdot (y_1^2 + y_2^2 + \cdots y_n^2)}$$

Taking square on both side.

$$(x_1 y_1 + x_2 y_2 + x_n y_n)^2 \leq (x_1^2 + x_2^2 + \cdots x_n^2) \cdot (y_1^2 + y_2^2 + \cdots y_n^2)$$

we can write $\left(\sum_{r=1}^{n} x_r y_r\right)^2$ ↑ we can write ↑ $\left(\sum_{r=1}^{n} x_r^2\right)\left(\sum_{r=1}^{n} y_r^2\right)$

(ii) Minkoski Inequality.

$$\sqrt{\sum_{r=1}^{n}(x_r+y_r)^2} \le \sqrt{\sum_{r=1}^{n} x_r^2} + \sqrt{\sum_{r=1}^{n} y_r^2}$$

$$(\vec{u}+\vec{v})(\bar{u}+\bar{v}) = \bar{u}\cdot\bar{u} + \bar{u}\cdot\bar{v} + \bar{v}\cdot\bar{v} + \bar{v}\cdot\bar{v}$$

$$|\bar{u}+\bar{v}|^2 = |\bar{u}|^2 + |\bar{v}|^2 + \bar{u}\cdot\bar{v} + \bar{u}\cdot\bar{v}$$

$$|\bar{u}+\bar{v}|^2 = |\bar{u}|^2 + |\vec{v}|^2 + 2|\bar{u}||\bar{v}|$$

$$|\vec{u}+\vec{v}|^2 = (|\vec{u}| + |\vec{v}|)^2$$

$$|\vec{u}+\vec{v}| \le |\vec{u}| + |\vec{v}|$$

$$\sqrt{(x_1+y_1)^2 + (x_2+y_2)^2 + \cdots (x_n+y_n)^2} \le \sqrt{x_1^2 + x_2^2 + \cdots x_n^2} + \left(\sqrt{y_1^2 + y_2^2 + \cdots y_n^2}\right)$$

Hence proved

Lec # 31 Mth405

If $x_1, x_2 \cdots x_n \in R$, then show that

(1) $\left(\sum_{k=1}^{n} x\right)^2 \leq n \sum_{k=1}^{n} x_k^2$

Proof We know that

$$\left(\sum_{r=1}^{n} \{x_r y_r\}\right)^2 \leq \sum_{r=1}^{n} x_r^2 \cdot \sum_{r=1}^{n} y_r^2$$

Put $r = k$ Then

$$\left(\sum_{k=1}^{n} x_k y_k\right)^2 \leq \sum_{k=1}^{n} x_k^2 \cdot \sum_{k=1}^{n} y_k^2$$

$y_k = 1$

$$\left(\sum_{k=1}^{n} x_k \cdot 1\right)^2 \leq \sum_{k=1}^{n} x_k^2 \cdot \sum_{k=1}^{n} (1)^2$$

$$\left(\sum_{k=1}^{n} x_k\right)^2 \leq \sum_{k=1}^{n} x_k^2 \cdot n,$$

$$\left(\sum_{k=1}^{n} x_k\right)^2 \leq n \sum_{k=1}^{n} x_k^2$$

Question: 2

$$(x_1 + x_2 + x_3 + x_4 + x_5)^2 \leq 5(x_1 + x_2 + x_3 + x_4 + x_5)$$

We know that:

$$\sum_{k=1}^{n} x_k^2 \leq n \sum_{k=1}^{n} x_k^2$$

Put $n = 5$

$$\sum_{k=1}^{5} x_k^2 \leq 5 \sum_{k=1}^{5} x_k^2$$

$$(x_1 + x_2 + x_3 + x_4 + x_5)^2 \leq 5(x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2)$$

Let $x(x_1, x_2, x_3, \ldots x_n), y(y_1, y_2, y_3, \ldots y_n) \in$

Show that

$d(x,y) = \sqrt{\sum_{j=1}^{n} (x_j - y_j)^2}$ is metric.

Sol

$$d(x,y) = \sqrt{(x_1-y_1)^2 + (x_2-y_2)^2 + \cdots + (x_n-y_n)^2}$$

m_1) non negative

$$d(x,y) \geq 0 \quad \forall\ x, y \in X \neq \phi$$

$$(x_1-y_1)^2 \geq 0 + (x_2-y_2)^2 \geq 0 \cdots + (x_n-y_n)^2 \geq 0$$

$$(x_1-y_1)^2 + (x_2-y_2)^2 + \cdots + (x_n-y_n)^2 \geq 0$$

$$\sqrt{(x_1-y_1)^2 + (x_2-y_2)^2 + \cdots + (x_n-y_n)^2} \geq 0$$

$$\sqrt{\sum_{j=1}^{n} (x_j-y_j)^2} \geq 0, \quad d(x,y) \geq 0$$

m_2) Reflexive property.

$$d(x,y) = 0 \qquad x = y$$

$$d(x,y) = 0, \quad \sqrt{\sum_{j=1}^{n} (x_j-y_j)^2} = 0$$

$$\sqrt{(x_1-y_1)^2 + (x_2-y_2)^2 + \cdots + (x_n-y_n)^2} = 0$$

$$(x_1-y_1)^2 + (x_2-y_2)^2 + \cdots + (x_n-y_n)^2 = 0$$

check all points

$(x_1-y_1)^2=0$	$(x_2-y_2)^2=0$	$(x_n-y_n)^2=0$
$x_1-y_1=0$	$x_2-y_2=0$	$x_n-y_n=0$
$x_1=y_1$	$x_2=y_2$	$x_n=y_n$

$(x_1, x_2 \dots x_n) = (y_1, y_2, \dots y_n)$

$x = y$

n3) Symmetric property

$$d(x,y) = d(y,x) \quad \forall\, x,y \in X \neq \phi$$

$$d(x,y) = \sqrt{\sum_{j=1}^{n}(x_j-y_j)^2} = \sqrt{\sum_{j=1}^{n}(-(y_j-x_j))^2} = \sqrt{\sum_{j=1}^{n}(y_j-$$

$$= d(y,x)$$

nu) Triangular property.

$$d(x,y) \leq d(x,z)+d(z,y) \quad \forall\, x,y,z \in X \neq \phi$$

$$d(x,y) = \sqrt{\sum_{j=1}^{n}(x_j-y_j)^2} = \sqrt{\sum_{j=1}^{n}((x_j-z_j+z_j-y_j))^2}$$

$$\sqrt{\sum_{j=1}^{n}((x_j-z_j)+(z_j-y_j))^2} \leq$$

$$\leq \sqrt{\sum_{j=1}^{n}(x_j-z_j)^2} + \sqrt{\sum_{j=1}^{n}(z_j-y_j)^2}$$

$$d(x,y) \leq d(x,z)+d(z,y)$$

Bounded function for given metric space (X,d).

If In which exists a positive real number $d(x,y) \leq n \quad \forall\, x,y \in X \neq \phi$ then (X,d) is bounded metric space.

e.g $d(x,y) = \begin{cases} 0 & \text{if } x=y \\ 1 & \text{if } x \neq y \end{cases}$ $\quad \frac{9}{1+a} < 1$

$$d'(x,y) = \frac{k\,d(x,y)}{1+n\,d(x,y)}$$

$$d''(x,y) = \min(1, d(x,y))$$

$$d''(x,y) \leq 1$$

If $\sup\{d(x,y) \;\forall\, y \in X\}$ is finite then unbounded metric space

e.g $\quad d(x,y) > n$

$$d(x,y) = |x-y|$$

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Short Lecture 32 Open sphere metric space
Virtual University
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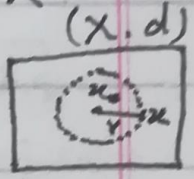
Lecture 32 MTH405

Open sphere and close sphere.

Let (X,d) be a metric space $r>0$ is positive real number x_0 for given point $x_0 \in X$

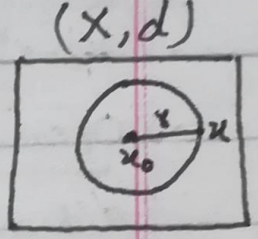


Open sphere

$$B_r(x_0) = S_r(x_0) = \{x \in X : d(x,x_0) < r\}$$

skin or boundary is not included where r = radius

x_0 = centre, B = Ball, S = sphere



Close sphere

$$Br[x_0] = S_r[x_0] = \{x \in X : d(x,x_0) \leq r\}$$

Skin or boundary is included.

Point

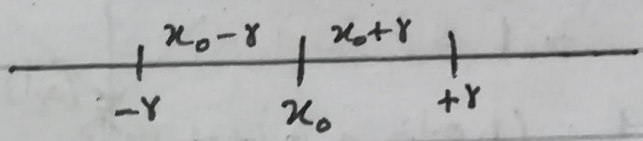
$d(x,x) = 0$ but $S_r[x] \neq \phi$, $S_r(x) \neq \phi$

Open sphere in $\mathbb{R}$ (Under usual metric)

$d(x,y) = |x-y|$

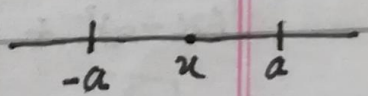
$$S_r(x_0) = \{x \in \mathbb{R} : d(x,x_0) < r\}$$
$$= \{x \in \mathbb{R} : |x-x_0| < r\}$$
$$-r < x - x_0 < r$$

add x_0 $\quad x_0 - r < x < x_0 + r$



$|x| = a$

$-a < x < a$



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Length of interval $(-r, r) = r-(-r) = r+r = 2r$

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$d(x,y) = \sqrt{|x-y|}$ $\quad y = x_0$

$S_r(x_0) = \{x \in \mathbb{R} : d(x,x_0) < r\}$

$= \{x \in \mathbb{R} : \sqrt{|x-x_0|} < r\}$

$2 < 3$

$2^2 < 3^2$

$4 < 9$

$\sqrt{|x-x_0|} < r$

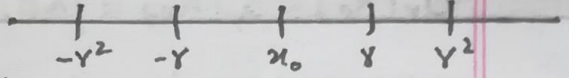
$|x-x_0| < r^2$

$-r^2 < x - x_0 < r^2$

add x_0

$x_0 - r^2 < x < x_0 + r^2$

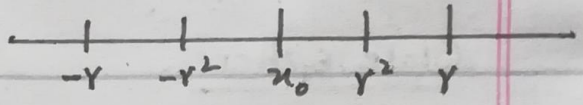
when $r > 1$



when $0 < r < 1$

$r = \frac{1}{2}$

$r^2 = \frac{1}{4}$



Open sphere on $\mathbb{R}^2$ (Taxicab metric space)

$d(P_1,P_2) = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$

let $P_1 = P = (x,y)$, $P_2 = P_0 = (0,0)$

$d(P,P_0) = \sqrt{(x-0)^2 + (y-0)^2}$

$= \sqrt{x^2 + y^2}$

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$S_r(P_0) = \{P \in \mathbb{R}^2 : d(P,P_0) < r\}$

$= \{P \in \mathbb{R}^2 : \sqrt{(x-0)^2 + (y-0)^2} < 1\} \because r = 1$

$\sqrt{(x-0)^2} + \sqrt{(y-0)^2} < 1$

$\sqrt{x^2} + \sqrt{y^2} < 1$

$|x| + |y| = 1$

$\pm x \pm y = 1$

$x + y = 1 \quad (1,0)$

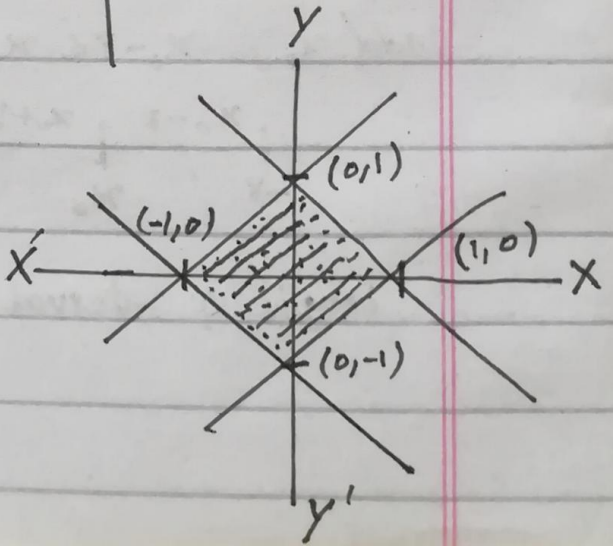
$x - y = 1 \quad (0,-1)$

$-x + y = 1 \quad (0,1)$

$-x - y = 1 \quad (-1,0)$

$\sqrt{x^2 + y^2} < \sqrt{x^2} + \sqrt{y^2}$

$\sqrt{x^2} = |x|$



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open sphere in Chebyshev metric on $\mathbb{R}^2$

$P_1(x_1, y_1)$, $P_2(x_2, y_2) \in \mathbb{R}^2$

$d(P_1, P_2) = \max\{|x_1 - x_2|, |y_1 - y_2|\}$

Let $P_1 = P = (x, y)$, $P_2 = P_0 = (0, 0)$

$S_r(P_0) = \{P \in \mathbb{R}^2 : d(P, P_0) < r\}$

$= \{P \in \mathbb{R}^2 \ \max\{|x-0|, |y-0|\} < 1\}$ $r = 1$

$\max\{|x|, |y|\} \leq 1$

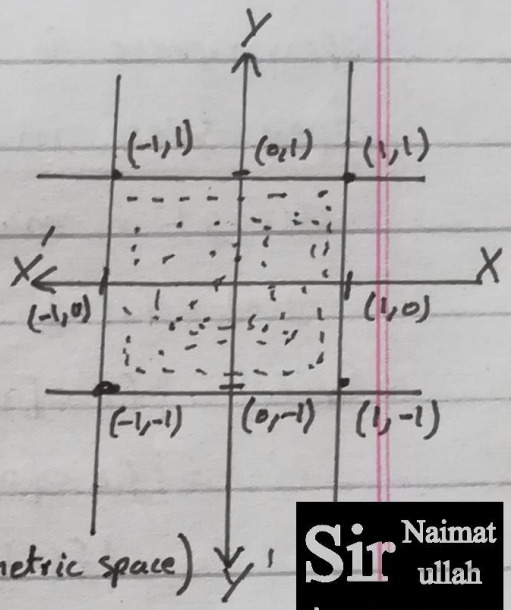
$\max\{\pm x, \pm y\} \leq 1$

$(1, 0) = 1$ $(0, -1) < 1$

$(0, 1) = 1$ $(-1, 0) < 1$

$(-1, 1) = 1$ $(1, 1) \leq 1$

$(1, -1) = 1$ $(-1, -1) < 1$



open spher in (discrete metric space)

$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$

$S_r(x_0) = \{x \in X : d(x, x_0) < r\}$ $x_0 = y$

case I $r \leq 1$

$S_r(x_0) = \{x \in X ; d(x, x_0) < 1\}$

$= \{x \in X ; d(x, x_0) = 0\}$ $x = x_0$

$= \{x_0\}$ only centre

case II $r > 1$

let $x \in X$ be an arbitrary point $x_0 \neq x$

(centre of open sphere)

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$x \in S_r(x_0)$, $U\{x\} \subseteq S_r(x_0)$

$S_r(x_0) \subseteq X$

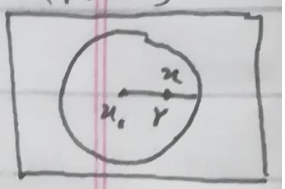
$U\{x\} \subseteq S_r(x_0) \subseteq X$

$\{x\} \cup \{x\} \cup \{x\} \cdots = X$

$X \subseteq S_r(x_0) \subseteq X$ $\quad A \subseteq B \subseteq A$, $A = B$

$S_r(x_0) = X$

(X, d)



open sphere is complete metric space when $r > 1$

Open spere in $c[a,b]$

$$d(f,g) = \max |f(x) - g(x)| \quad , \forall x \in [a,b]$$

$g(x) = f_0(x)$ $\qquad r = 1$

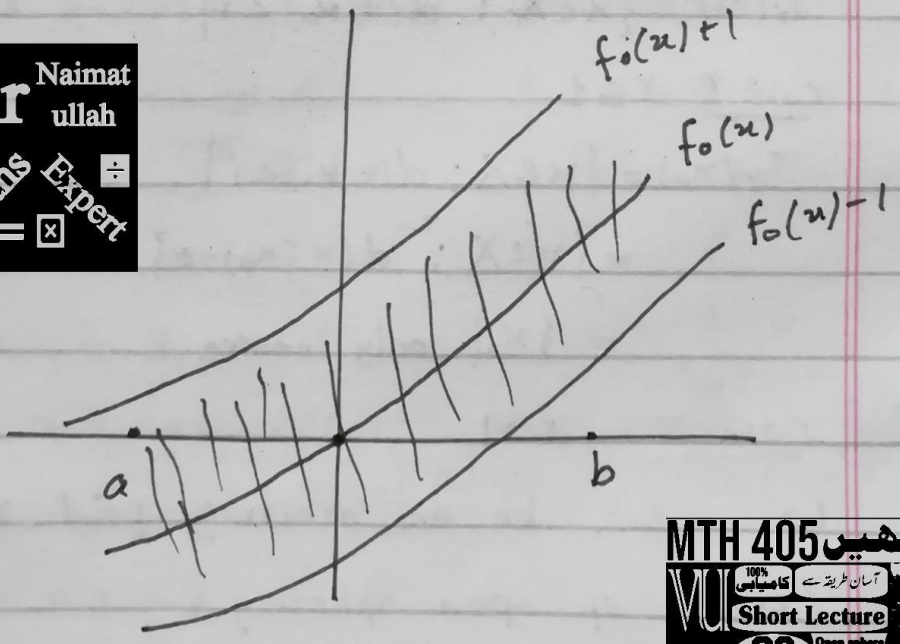
$$S_r(f_0) = \{f \in c[a,b] \; ; \; d(f, f_0) < 1\}$$

$$= \{f \in c[a,b] \quad \max |f(x) - f_0(x)| < 1\}$$

$$|f(x) - f_0(x)| < 1$$

$$-1 < f(x) - f_0(x) < 1$$

add $f_0(x)$, $f_0(x) - 1 < f(x) < f_0(x) + 1$



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Open set in metric space

Let $(X.d)$ be a metric space $A \neq \phi$ such That $A \subseteq X$, if $\forall\, x \in A$ and There exist $r>0$ suh That $S_r(x) \in A$, Then 'A' is open set.

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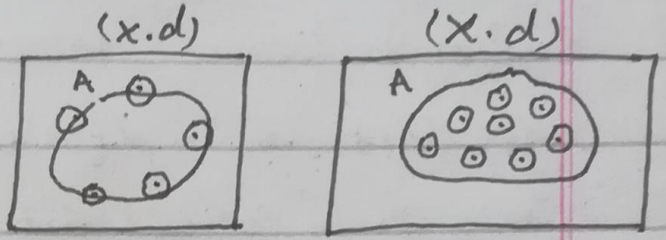
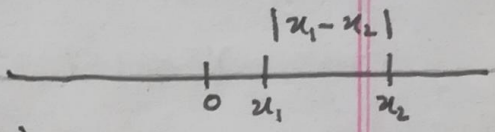


not open set open set

1 Usual metric space

$d(x_1, x_2) = |x_1 - x_2|$

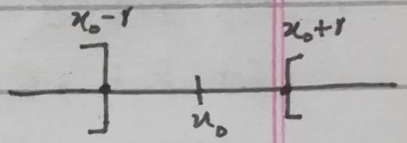


$S_r(x_0) = \{x \in \mathbb{R} : d(x, x_0) < r\}$

$= \{x \in R : |x - x_0| < r\}$ $x_1 = x$, $x_2 = x_0$

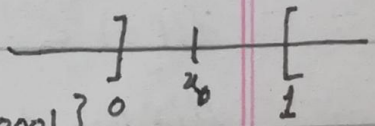
$= \{x \in R : -r < x - x_0 < r\}$

Add x_0, $x_0 - r < x < x_0 + r$



put $x_0 = 0.999$, $r = 0.0001 > 0$

$S_r(x_0) = \{x \in R : x_0 - r < x < x_0 + r$



$S_{0.0001}(0.999) = \{x \in R\ 0.999 - 0.0001 < x < 0.999 + 0.0001\}$

$= \{x \in R : 0.9989 < x < 0.9991\}$ ~~Open Set~~

$]0.9989, 0.9991[\subseteq]0\ 1[$ which hold $\forall$ $x \in]0\ 1[$ as well $]0\ 1[$ is open set in $(\mathbb{R}.d)$

② open set in $\mathbb{R}^2$ (under usual metric space)

Prove That The set $U = \{(x,y) \in R^2 : x^2 + y^2 < 1\}$ is an open set in $(d, \mathbb{R}^2)$

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Proof

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$$U = \{(x,y) \in R^2 : x^2 + y^2 < 1\}$$

$$= \{(x,y) \in R^2 : (x-0)^2 + (y-0)^2 < 1\}$$

$$= \{(x,y) \in R^2 : \sqrt{(x-0)^2 + (y-0)^2} < 1\}$$

$x < y$
$\sqrt{x} < \sqrt{y}$

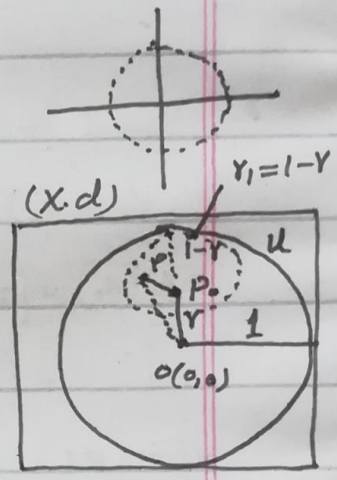
$d(P_1, P_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ is a Euclidean distance $P_1(x_1, y_1)$, $P_2(x_2, y_2)$

$$U = \{(x,y) \in R^2 : d[(x,y),(0,0)] < 1\}$$

Taking P_0 in 'U' such that

$d(P_0, 0) = r$

$r_1 = 1 - r$



$x \in A$, $x \in B$
$A \subseteq B$

Now prove that $S_{r_1}(P_0) \subseteq U$

let $P \in S_{r_1}(P_0)$ — 1

an applying triangular inequality for $P, P_0, O(0,0)$

$d(P,0) < d(P,P_0) + d(P_0,0)$

$d(P,0) < r_1 + r = 1 - r + r = 1$

$d(P,0) < 1$, $P \in U$ — ②

from ① and ② $S_{r_1}(P_0) \subseteq U$ which holds arbitrary for all points of 'U'

So, 'U' is a open set in (R^2, d)

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Clopen Theorem

In a metric space $(X.d)$, the empty set ϕ and the full space 'X' are open sets.

Proof **Case 1** $\phi = \{x, x \neq x_0\}$

let "ϕ" is not an empty set in $(X.d)$. There exist a point say $x \in \phi$ and $r>0$; $S_r(x) \nsubseteq \phi$, but ϕ is empty and does not have any point 'x" a contradiction occurs "ϕ" is an empty set and is an open set in $(X.d)$

Case II 'X' is open set

$\forall\ x \in X$ and There exist $r>0$ such That $S_r(x) \subseteq X$, so, 'X' is an open set.

Deductions

if 'A' is open set in $(X.d)$ Then A' is closed.

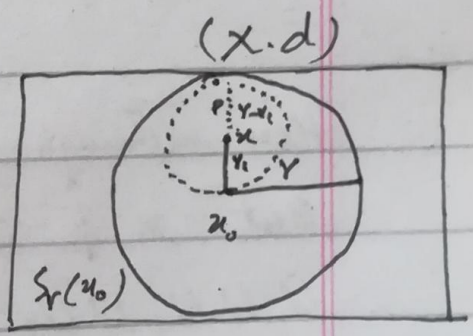
if 'ϕ' is open set in $(X.d)$ Then ϕ' is closed. $\phi' = X$

if 'X' is open set in $(X.d)$ Then X' is closed $X' = \phi$

Open sphere for any metric space

each open sphere is any metric space is an open set. However, its converse is not true.

Proof Let $S_r(x_0)$ be an open sphere in $(X.d)$.



we are to prove

$\forall\ x \in S_r(x_0)$, and There exist $r>0$, such That $S_p(x) \subseteq S_r(x_0)$

taking $x \in S_r(x_0)$ such That

$d(x, x_0) = r_1 \rightarrow (1)$, $p = r - r_1 \rightarrow (2)$

Now draw an open sphere at $x \in X$ of size

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$p = r - r_1$ $S_p(x)$ such that if $y \in S_p(x)$ Then

$d(y,x) < p$ — ③

Now applying triangular inequality for, $x, y, x_0 \in X$

$d(y,x_0) < d(y,x) + d(x,x_0)$

$< p + r_1$, $= r - r_1 + r_1 = r$

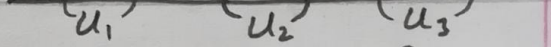
$d(y,x_0) < r$, so, $y \in S_r(x_0)$

$y \in S_p(x)$ and $y \in S_r(x_0)$

So, $S_p(x) \subseteq S_r(x_0)$ $\quad x \in A, x \in B, A \subseteq B$

which holds $\forall\ x \in S_r(x_0)$ is an open set in (X, d)

Conversly

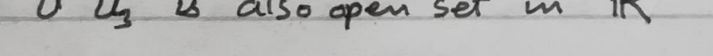


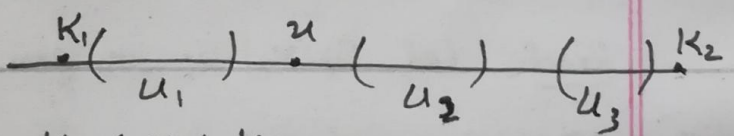
Let U_1, U_2, U_3 are open sets in $\mathbb{R}$

$U_1 \cup U_2 \cup U_3$ is also open set in $\mathbb{R}$

But it is not open interval

{ if $(a,b) \in R$ such That $a < x < b \Rightarrow x \in (a,b)$





Here although $K_1 < x < K_2$

But $x \notin U_1 \cup U_2 \cup U_3$

$U_1 \cup U_2 \cup U_3$ is not an open sphere in $\mathbb{R}$

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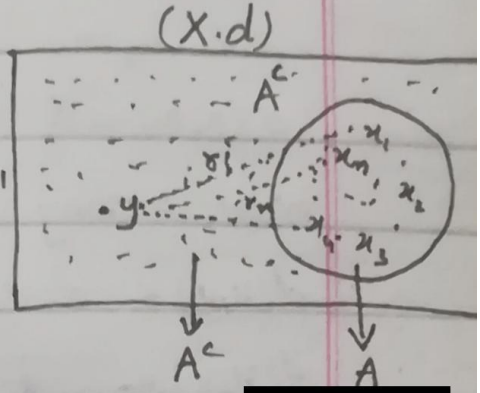
Theorem; The compliment of finite set in a metric space is open set;



Proof Let $A=\{x_1, x_2, \dots x_n\}=\{x_i\}_{i=1}^{n}$ be a finite subset of "X"

$A \subseteq X$

Let $y \in A^c$, $y \notin A$

$y \neq x_i \ (x_1, x_2, \dots x_n) \quad i = 1, 2, \dots n$

$d(x_i, y) \neq 0$

Let $d(x_i, y) = r_i \quad \forall\ i = 1, 2, 3, \dots n$ and also

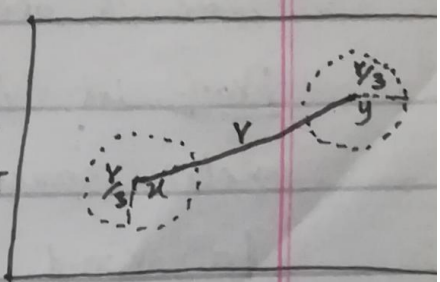
Let $r_i = \min\{r_1, r_2 \dots r_n\} = \{r_i\}_{i=1}^{n}$

Let an open sphere of radius "r" with center at "y" does not contain an point of set "A" so,

$S_r(y) \cap A = \phi$ and $A^c \cap A = \phi$ So, $S_r(y) \subseteq A^c$

it holds for all $y \in A^c$, A^c is an open set in $(X.d)$

Housdorff property every Two distinct points in a metric space are separated by pair of disjoint open sphere containing the respective points.



Proof let $(X.d)$ be an arbitrary metric space containing Two distinct Points. x and y such That $d(x,y) = r$, Now taking Two open sphere at x and y of radius : $\frac{r}{3}$:

we are to prove That $S_{r/3}(x) \cap S_{r/3}(y) = \phi$

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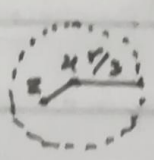
Let if $S_{r/3}(x) \cap S_{r/3}(y) \neq \phi$ There exist z pint

$z \in S_{r/3}(x) \cap S_{r/3}(y)$

$z \in S_{r/3}(x)$, $z \in S_{r/3}(y)$



$d(x,z) < r/3$, $d(y,z) < r/3$

Now appling Trinangular inequality on x, y, z

$d(x,y) \leq d(x,z) + d(y,z)$

$d(x,y) < \frac{r}{3} + \frac{r}{3}$

$r < \frac{r+r}{3}$, $r < \frac{2r}{3}$, $1 < \frac{2}{3}$ is false

So, our sopposition $S_{\frac{r}{3}}(x) \cap S_{\frac{r}{3}}(y) \neq \phi$ is wrong

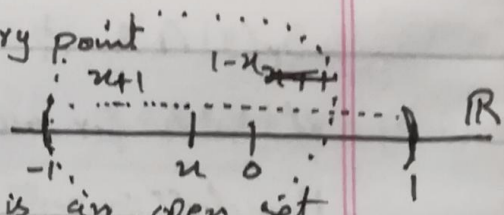
Hence $S_{\frac{r}{3}}(x) \cap S_{r/3}(y) = \phi$. x, y distinct point open sphere are dijoint pair.

Theorem Any open interval on $\mathbb{R}$ is an open set

Proof Let $(-1,1) \subseteq \mathbb{R}$ be an open interval in $\mathbb{R}$.

$x \in (-1,1)$ be its an arbitrary point

Taking $r = \min\{|x+1|, |1-x|\}$



$S_r(x) \subseteq (-1,1)$, so, $(-1,1)$ is an open set

Theorem Non-empty subset in a discrete metric space is open set. $d(x,y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$

Proof Let $A \neq \phi$ bet a subset of $\mathbb{R}$ (x,d) a discrete metric space

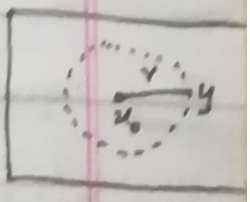
for $r \geq 1$, $S_r(x) = A = X \quad \forall x \in X = A$

for $r < 1 \quad S_r(x) = \{x\} \quad \forall x \in A = X$

from $r \geq 1$ and $r < 1$ is a open set in metric space.

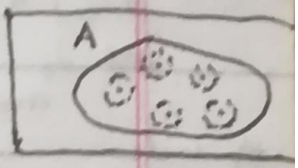
Land Mark Theorem for open sets (Arbitrary Union)

Arbitrary union of open set in a metric space is open set.



Proof open sphere, $S_r(x) = \{y \in x : d(x,y) < r\}$

open set 'A' is open in $(X.d)$ if all of its centre of some open sphere



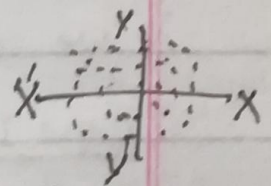
Arbitrary Union

finite — countable — infinit — Uncountable

$A_1 \cup A_2 \cup A_3 \cdots A_n$ $\quad A_1 \cup A_2 \cup A_3 \cdots = \bigcup_{i=1}^{\infty} A_i$ $\quad \bigcup_{\alpha \in I} A_\alpha$

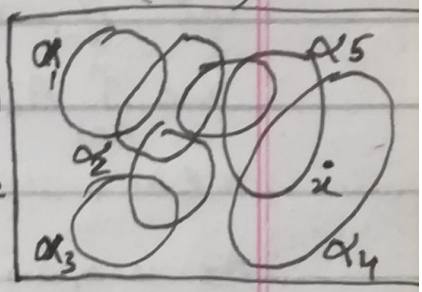
e.g uncountable $A_x = \{(x,y) : y \in R\}$

$x \in R$ $\quad A_{x \in R} = \{(x,y) : y \in R\}$



Union of $(\bigcup A_{x \in R}) = \bigcup \{(x,y) : y \in R\} = R \times R = XY\text{-Plane}$ $(X.d)$

let $\{A_\alpha\}_{\alpha \in I}$ be the arbitrary collection in metric space $(X.d)$



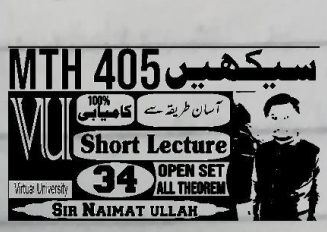
1 Then if $\bigcup_{\alpha \in I} A_\alpha = \phi$ Then by define "ϕ" is open set in $(X.d)$

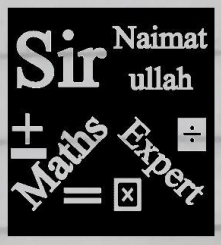
② If $\bigcup_{\alpha \in I} A_\alpha \neq \phi$, There exist $x \in A_\alpha$ for some $\alpha \in I$

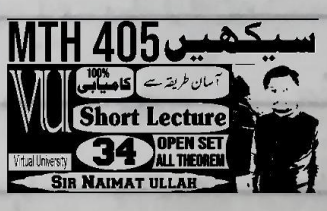
A_α is open set in $(X.d)$, There exist (open sphere $r > 0$ at 'x') such That $S_r(x) \subseteq A_\alpha$ for some $\alpha \in I$

$S_r(x) \subseteq A_\alpha \subseteq \bigcup_{\alpha \in I} A_\alpha$ as x is arbitrary ~~function~~

$\bigcup_{\alpha \in I} A_\alpha$ is open set in $(X.d)$







MTH 405 سیکھیں
VU 100% کامیابی آسان طریقے سے
Short Lecture 34 OPEN SET ALL THEOREM
Virtual University
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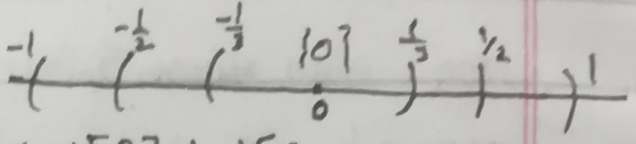
Land Mark Theorem for open sets (Finite intersection)

Finite intersection of open sets in a metric space is open set

Proof infinite



$\bigcap_{n=1}^{\infty} \left]-\frac{1}{n}, \frac{1}{n}\right[= \left]-1, 1\right[\cap \left]-\frac{1}{2}, \frac{1}{2}\right[\cap \left]-\frac{1}{3}, \frac{1}{3}\right[\cap \ldots\ldots \infty$

If $\lim_{n\to\infty} \bigcap_{n=\infty}^{\infty} \left]-\frac{1}{n}, \frac{1}{n}\right[= \{0\}$ it is singleton set is a close set

Finite

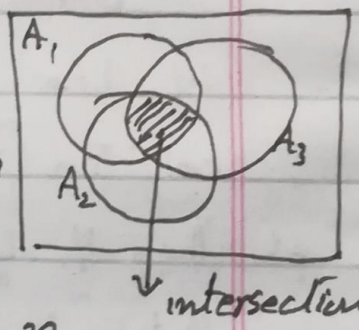
Let $\{A_1, A_2, \ldots A_n\} = \{A_\alpha\}_{\alpha=1}^{n}$ be the finite collection of open set

we are to prove $A_1 \cap A_2 \cap \cdots A_n = \bigcap_{\alpha=1}^{n} A_\alpha$ is open

Case I if $\bigcap_{\alpha=1}^{n} A_\alpha = \phi, X$, which is open sets

Case II

Let $\bigcap_{\alpha=1}^{n} A_\alpha \neq \phi$ Then



$x \in \bigcap_{\alpha=1}^{n} A_\alpha$, so, $x \in A_\alpha$, $\alpha = 1, 2, \ldots n$

$x \in A_1$, $x \in A_2 \ldots\ldots x \in A_\alpha$

A_α is open set $\forall\ \alpha = 1, 2, 3, \ldots n$

There exist $r_1, r_2 \ldots\ldots r_n > 0$ such that

$S_{r_1}(x) \subseteq A_1$, $S_{r_2}(x) \subseteq A_2 \ldots\ldots S_{r_n}(x) \subseteq A_n$

Let $r = \min\{r_1, r_2, \ldots r_n\}$

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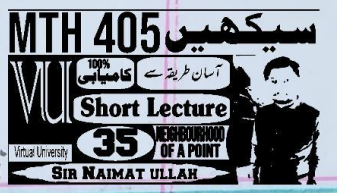
$S_r(x) \subseteq \{A_1, A_2, \ldots A_n\}$

$S_r(x) \subseteq A_1 \cap A_2 \cap \ldots \cap A_n$

$S_r(x) \subseteq \bigcap_{\alpha=1}^{n} A_\alpha$, where x is arbitrary point

So, $\bigcap_{\alpha=1}^{n} A_\alpha$ finite Intersection is open set in metric space

Naimat ullah

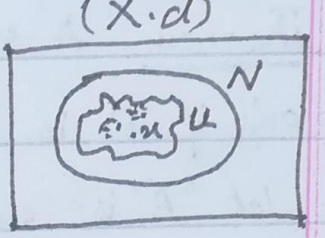


MTH 405 Lectur 35 Nomi

Neighbourhood of a point

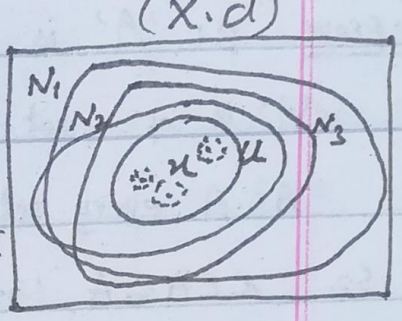
Neighbourhood means subset: ■ A subset

'N' is said to be The Neighbourhood of a poin 'x' If There exist an open set 'u' containing such That $x \in u \subseteq N$ stand for Nghd $(x) \to N_x$

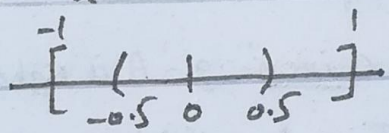


Neighbourhood system of Point

If $x \in u \subseteq N_1, N_2, N_3 \ldots = \{N_i\}_{i \in I}$ is said to be Neighbourhood system of point 'x'



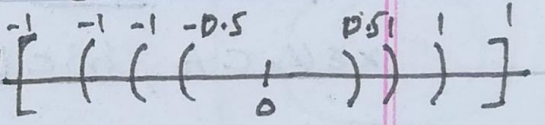
e.g on $\mathbb{R}$



$0 \in (-0.5, 0.5) \subseteq [-1,1]$

So, $[-1,1]$ is Nghd of 0

$0 \in (-0.5, 0.5) \subseteq [-1,1), (-1,1), (-1,1]$

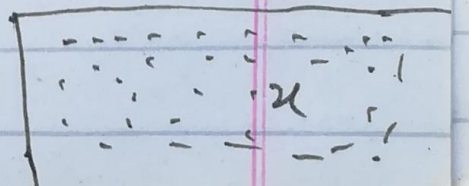


are Nghd system of '0'

eg In discrete metric space every subset containg 'x' is Nghd of 'x' (x, d_0), we have $A \subseteq X$ such That $x \in A$, means $x \in A \subseteq X$ (X_x) Nghd of 'x' in (x, d_0), every singleton set is open $x \in \{x\} \subseteq A$ A is Nghd of 'x' $\forall\ x \in A$, A is Nghd of all its points.

Theorem the metric space itself is a Nghd of each of its point.

$x \in X \subseteq X$



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Short Lecture 35 NEIGHBOURHOOD OF A POINT
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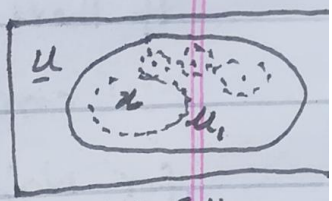
Proof In a metric space $(X.d)$, let 'x' be an arbitrary point, Then $x \in X \subseteq X$, X is open set and X is also Nghd of 'x'

Theorem A subset of metric space is open if and only if it is the Neighbourhood of each of its point

Proof let 'A' is subset of 'X'

Given ⇒ P ; A is an open set



$x \in U_1 \subseteq U$

U_1 is open set

U is Nghd of x

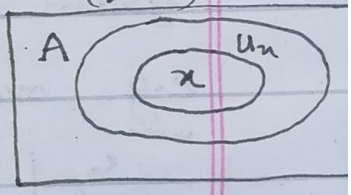
To prove Q: 'A' is Nghd of each of its point

'A' is open set in X, $\forall x \in A$ — I

$A \subseteq A$ every set is subset of itself — II

So, $x \in A \subseteq A$, 'x' is arbitrary point of 'A' & 'A' is Nghd of all its point.

$(X.d)$



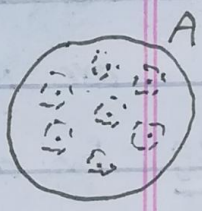
$x \in B$, $\{x\} \subseteq B$

Conversly Given : Q: A is Nghd of each its point. **To proive** : 'A' is an open set

$x \in U_x \subseteq A$, $\{x\} \subseteq U_x \subseteq A$

Taking Union for all such 'x'

$\bigcup_{x \in A} \{x\} \subseteq \bigcup_{x \in A} U_x \subseteq A$; $\bigcup_{x \in A} U_x$ is union of open set



$A \subseteq B \subset A$, $A = B$

$$A \subseteq \bigcup_{x \in A} U_x \subseteq A$$

$$A = \bigcup_{x \in A} U_x$$

Hence A is open set, being arbitrary union of open set

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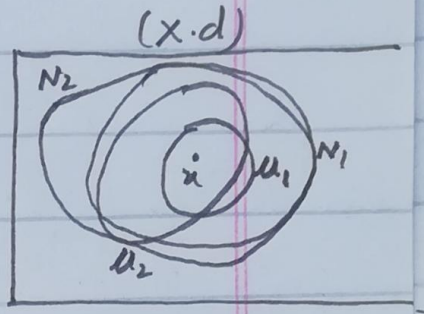
MTH 405 Lecture 36 Nomi

Theorem Finite intersection of the Neighbourhood of a point, is also its Neighbourhood.



Proof let $N_1, N_2, \dots N_\gamma = \{N_j\}_{j=1}^{\gamma}$ be the Nghd system point 'x' in $(X.d)$

there exist $U_1, U_2, \dots U_\gamma = \{U_j\}_{j=1}^{\gamma}$

the collection of open sets

$x \in U_1 \subseteq N_1$, $x \in U_2 \subseteq N_2$, $\dots$ $x \in U_j \subseteq N_j$, $j = 1, 2, 3, \dots \gamma$

$x \in U_1 \cap U_2 \cap \dots \cap U_\gamma \subseteq N_1 \cap N_2 \cap \dots \cap N_\gamma$

$x \in A$, $x \in B$

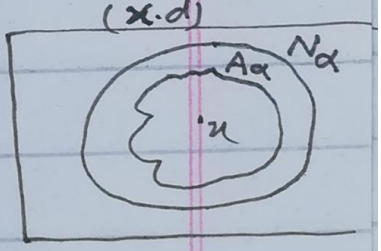
$x \in A \cap B$

$$x \in \bigcap_{j=1}^{\gamma} U_j \subseteq \bigcap_{j=1}^{\gamma} N_j$$

$\bigcap_{j=1}^{\gamma} U_j$ = finit intersection of open set so,

U_j = is an open set and $\bigcap_{j=1}^{\gamma} N_j$ is Nghd of 'x'

Theorem Arbitrary union Neighbourhood of a point is also its Neighbourhood.



Proof let $\{N_\alpha\}_{\alpha \in I}$ be The arbitrary collection of Nghd of point 'x' in $(X.d)$

each N_α is Nghd of 'x' where $\forall \alpha \in I$

There exist an open set say A_α such that $x \in A_\alpha \subseteq N_\alpha$

$\forall \alpha \in I$ So, $x \in \bigcup_{\alpha \in I} A_\alpha \subseteq \bigcup_{\alpha \in I} N_\alpha$

$x \in A_1$

$x \in A_1 \cup A_2 \cup \dots$

$\bigcup_{\alpha \in I} A_\alpha$ being arbitrary union of open sets.

So, $\bigcup_{\alpha \in I} N_\alpha$ is Nghd of 'x'

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Sir Naimat ullah Maths Expert
MTH 405 سیکھیں
VU 100% کامیابی آسان طریقے سے Short Lecture 37,38 LIMIT POINT DERIVED SETS
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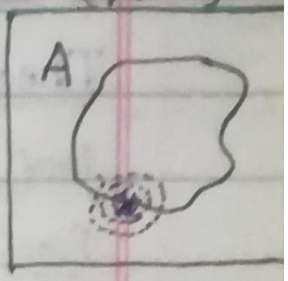
MTH 405 lecture 37 Nomi (X.d)



Limit point of set In a metric space (X.d). Let $A \subseteq X$ and any point $x \in X$ is said to be the limit point of 'A' if every open sphere at 'x' contain at least one point of 'A' other than 'x'

$$S_\delta(x) \cap A - \{x\} \neq \phi \quad \text{where } r>0$$

e.g; 0,1 are the limit points of $]0,1[$ because any open interval at $0,1 \notin]0,1[$

'0' and '1' contain at least one point of $]0,1[$

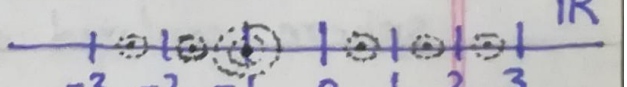
$S_r(0) \cap]0,1[- \{0\} \neq \phi$, $S_r(1) \cap]0,1[- \{1\} \neq \phi$

$(0-\varepsilon, 0+\varepsilon) \cap]0,1[- \{0\} \neq \phi$, $(1-\varepsilon, 1+\varepsilon) \cap]0,1[- \{1\} \neq \phi$

$(x-\varepsilon, x+\varepsilon) \cap]0,1[- \{x\} \neq \phi \quad \forall \varepsilon>0 , \forall x \in]0,1[$

e.g - Limit points of The set of integers $\mathbb{Z} \subseteq \mathbb{R}$

Sol $\mathbb{Z} = \{0, \pm1, \pm2, \pm3, \cdots\}$



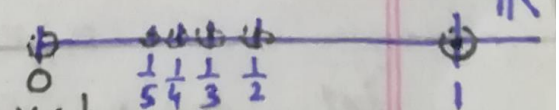
$\forall x \in \mathbb{R}$ There exist $r>0$

Such that $S_r(x) \cap \mathbb{Z} - \{x\} = \phi$

$\mathbb{Z}$ has no limit point because all integer's number's has some distance between each other

e.g; Limit point of $\{\frac{1}{n}\}_{n\in\mathbb{N}}$: $\lim_{n\to\infty} [\frac{1}{\infty}] = \{0\}$

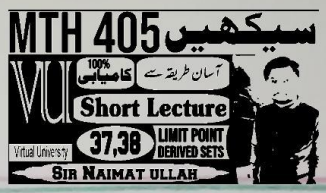
$\{\frac{1}{n}\}_{n\in\mathbb{N}} = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \cdots\}$



So, $\frac{1}{2} \notin S_{\frac{1}{2}}(1)$, $r<\frac{1}{2}$

By induction $\frac{1}{n+1} \notin S_{\frac{1}{n+1}}(0)$ if $n \in \mathbb{N}$ Then $n+1 \in \mathbb{N}$

0, is a limit point of $\{\frac{1}{n}\}_{n\in\mathbb{N}}$



Closed set / Derived set

A subset 'A' of metric space (X,d) $A \subseteq X$ is closed if it contains every limit point of itself.

The set of all limit points of A is called The derived set of A and denoted by A', A^c, A^d

e.g $0,1 \notin]0,1[$ $\quad A' = A^c = A^d =]0,1[^d = [0,1]$ is closed set

$\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$, $\mathbb{Z}^d = \{\} = \phi$

$0 \notin \{\frac{1}{n}\}_{n \in N} = \{1, \frac{1}{2}, \frac{1}{3}, \dots\}$, $\{\frac{1}{n}\}^d_{n \in N} = \{0\}$

Derived set of Rational (Q) and Irrational (Q')

Rational $(Q) = \{x = \frac{p}{q}, q \neq 0, p, q \in Z \subset R\}$

Irrational $(Q') = \{x \neq \frac{p}{q}, q \neq 0, p, q \in Z \subset R\}$

e.g Q: $\frac{1}{2}$, 7 , $\frac{7}{2}$, e.g $Q' = \pi, \sqrt{29}, e$

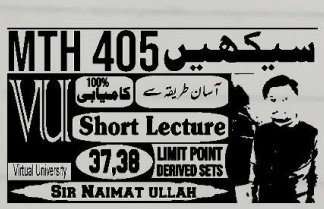
Q is infinite on $\mathbb{R}$ line, Q' is infinite on $\mathbb{R}$ Line

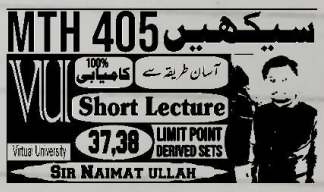
either $x \in Q$ or $x \in Q'$

$S_r(x) \cap Q - \{x\} \neq \phi$ when $x \in Q$

$S_r(x) \cap Q' - \{x\} \neq \phi$ when $x \in Q'$

So, derived set of $= Q^d = (Q^c)^d = \mathbb{R}$





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37,38 LIMIT POINT DERIVED SETS
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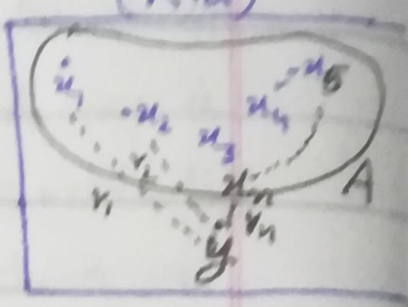
MTH-405 Lecture-38 Nomi

Theorem Finite subset of a metric space has no limit point.



Proof let $A \neq \phi$, $A \subseteq X$ be a subset of 'X' such that

$A = \{x_i\}_{i=1}^{n} = \{x_1, x_2, \ldots x_n\}$

let y is limit point of A, $S_r(y) \cap A - \{x\} = \{y\}$

Let $d(x_1, y) = r_1$, $d(x_2, y) = r_2 \ldots d(x_n, y) = r_n$

$r = \min\{r_1, r_2, \ldots r_n\} = \min\{r_i\}_{i=1}^{n}$

$S_r(y) = \{y \in X \; d(y, x_n) < r_n\}$

So, $S_r(y)$ has no point of A, other than 'y'

A has no limit point in 'X'

Theorem If 'x' is a limit point of a subset 'A' of a metric space (X.d) Then every open sphere centered at 'x' contains infinite number of points of 'A' different from 'x'.

Theorem The derived set of a non-empty set of discrete metric space is empty.

Proof $d_0 : X_0 \times X_0 \longrightarrow \mathbb{R}$; $d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$

$A \subseteq X$, $A^d = \{x \in X ; x \text{ is limit point of } A\}$

$= \{x \in X ; S_r(x) \cap A - \{x\} \neq \phi\}$

Let, $A \neq \phi$, $A \subseteq X$ and $A^d \neq \phi$ There exist $x : x \in A^d$, x is a limit point of A.

$\exists \; r > 0$ such that $S_r(x)$ should at least contain

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one point of 'A' other Than 'x'

if $r<1$ Then, $S_r(x)=\{y\in X_0 : d(y,x)<1\}$

$=\{y\in X_0 : d(y,x)=0\}$

$=\{x\}$ $\quad y=x$

$x\notin A^d$, 'x' is not a limit point of 'A'

$A\cap S_r(x)-\{x\}=\phi$, Hence, $A^d=\phi$.

Set Theoretic operations in Derived sets:

For any non-empty subset 'A' and 'B' of a metric space (X,d)

(i) if $A\subseteq B$ Then $A^d\subseteq B^d$ (ii) $(A\cap B)^d\subseteq A^d\cap B^d$

(iii) $(A\cup B)^d=A^d\cup B^d$

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Proof (i) $A^d=\{x\in X : A\cap S_r(x)-\{x\}\neq\phi, \forall r>0\}$

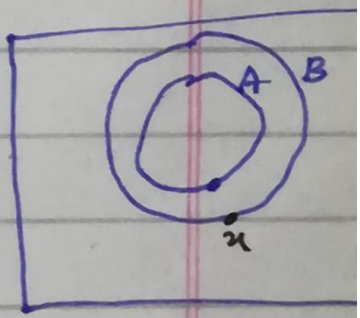
give $A\subseteq B$ and to prove $A^d\subseteq B^d$

Let $x\in A^d$; $A\cap S_r(x)-\{x\}\neq\phi \quad \forall r>0$ —1

$B\cap S_r(x)-\{x\}\neq\phi \quad \because A\subseteq B$

So, x is a limit point of 'B' as well $x\in B^d$ — II

So, $A^d\subseteq B^d$ from I and II



(ii) $(A\cap B)^d=A^d\cap B^d$

$A\cap B\subseteq A\ \wedge,\ A\cap B\subseteq B$

$(A\cap B)^d\subseteq A^d\ \wedge,\ (A\cap B)^d\subseteq B^d$

$(A\cap B)^d\subseteq A^d\cap B^d$

Counter example $\quad A^d\cap B^d=]1,2[$

$(A\cap B)^d=]2,3[$

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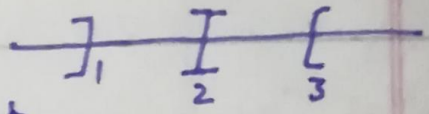
$]1,2[^d = [1,2]$, $]2,3[^d = [2,3]$

$]1,2[^d \cap]2,3[^d = [1,2] \cap [2,3] = \{2\}$

$]1,2[\cap]2,3[= \phi$



$(]1,2[\cap]2,3[)^d = \phi^d = \phi$

$Q \cap Q^c = \phi$ $\quad \phi \neq \{2\}$

$\mathbb{R} \cap \mathbb{R} = \mathbb{R}$ $\quad \phi \neq \mathbb{R}$

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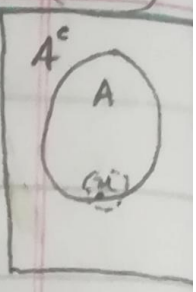
MTH-405 lecture 39 Nomi (X.d)



Closed set A subset 'A' of a metric space $(X.d)$, $A \subseteq X$ is closed set if it contains every limit point of itself

e.g '0' is the limit point of $\{\frac{1}{n}\}_{n=1}^{\infty}$ but '0' $\notin \{\frac{1}{n}\}_{n=1}^{\infty}$

So, it is not closed set.

e.g '0' and '1' are the limit point of open interval $]0,1[$ ~~and~~ but $0,1 \notin]0,1[$ is not closed set.

e.g '0' and '1' are the limit point of ~~open~~ close interval $[0,1]$ and $0,1 \in [0,1]$ is closed set.

e.g $R = Q \cup Q^c$, $Q^d = R = Q \cup Q^c$, $(Q^c)^d = R = Q \cup Q^c$

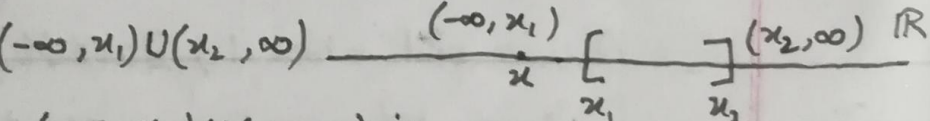
if $x \in Q^c$, then its limit point is Q and $x \notin Q$

if $x \in Q$, then its limit point is Q^c and $x \notin Q^c$

So, Q and Q^c are not closed in $\mathbb{R}$

Example close interval on $\mathbb{R}$ line are closed set

$[x_1, x_2]^c = (-\infty, x_1) \cup (x_2, \infty)$



we porove $(-\infty, x_1) \cup (x_2, \infty)$ is open

Let $x \in (-\infty, x_1) \cup (x_2, \infty)$, $x \in (-\infty, x_1)$ or $x \in (x_2, \infty)$

if $x \in (-\infty, x_1)$ and taking $r = |x_1 - x| > 0$ such that

$S_r(x) = \{y \in \mathbb{R} : d(x,y) < r\} =]x, y[\subseteq (-\infty, x_1)$

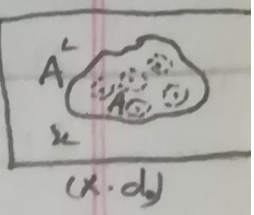
$S_r(x) \subseteq (-\infty, x_1)$, $S_r(x) \subseteq (-\infty, x_1) \cup (x_2, \infty)$

So, $(-\infty, x_1) \cup (x_2, \infty)$ is open set

$[(-\infty, x_1) \cup (x_2, \infty)]^c = [x_1, x_2]$ is closed set.

Example every subset of discrete metric space is closed set.

$$d_0(x,y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$



let $\phi \neq A \subseteq (X, d_0)$, To prove 'A' is closed set and 'A^c' is open set

let $x \in A^c$, for $r=1$, Taking open sphere $r=1$ at

$x \in A^c$, $S_r(x) = \{y \in X, d_0(x,y) < 1\}$

$= \{y \in X, d_0(x,y) = 0\}$

$= \{x\}$, $x = y$

$\{x\} \subseteq A^c$, $S_r(x) \subseteq A^c$ as 'x' is arbitrary point of A^c, so, A^c is open set, $(A^c)^c = A$ is closed set

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Theorem The Union of finite metric / number of closed sets is closed.

Proof Finite collection of closed sets $= \{A_1, A_2, \ldots A_n\} = \{A_i\}_{i=1}^n$

$\{A_1^c, A_2^c, \ldots A_n^c\} = \{A_i^c\}_{i=1}^n$ is open set | Finite intersection

$A_1^c \cap A_2^c, \ldots \cap A_n^c\} = \bigcap_{i=1}^n A_i^c$ is open sets | of open set is

$\{A^c \cap B^c = (A \cup B)^c$ de morgan's Law | open

$\{A_1^c \cap A_2^c \ldots \cap A_n^c\} = \bigcap_{i=1}^n A_i^c = \{A_1 \cup A_2 \cup \ldots \cup A_n\}^c$ | $= \left(\bigcup_{i=1}^n A_i\right)^c$

is open, Then $\left[\left(\bigcup_{i=1}^n A_i\right)^c\right]^c = \bigcup_{i=1}^n A_i$ is closed set

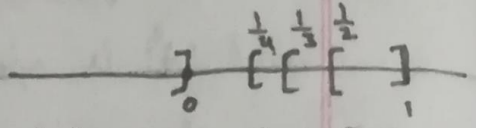
Infinitely

Let $A_n = \left[\frac{1}{n}, 1\right]$

$A_1 = [1,1]$, $A_2 = \left[\frac{1}{2}, 1\right]$, ...

as $n \to \infty$ Then $\frac{1}{n} = 0$, $A_n = (0,1]$ as $n \to \infty$

$A_1 \cup A_2 \cup \dots A_\infty = (0,1]$ is not closed sets



Theorem the Union of [redacted] arbitrary number of closed sets is closed.

Theorem The [redacted] intersection of finite number of closed set is closed.

Closeness of a finite subset of 'A' metric space.

Theorem; a finite subset of a metric space is closed set.

Proof Let $\phi \neq A \subseteq X$ be a non-empty subset of X

P; 'A' is finite $\quad Q$; 'A' is closed

اگر زیرِ کھائیں گے تو مر جائیں گے $= P \to Q$

مریں گے نہیں اگر زیر نہیں کھائیں گے $= \sim Q \to \sim P$

$P \to Q \equiv \sim Q \to \sim P$

$\sim Q$; 'A' is not closed set

$\sim P$ = 'A' is infinite

$\sim Q$ = 'A' is not closed set

There exist $x \in X : x \in A^d$ but $x \notin A$

$x \in A^d$, 'x' is a limit point of 'A'

every open sphere at 'x' will contain infinite many points of 'A'

So, 'A' is infinit ($\sim P$)

So, $\sim Q \to \sim P = P \to Q$

Theorem In a metric space a subset is closed set if and only if its compliment is open.

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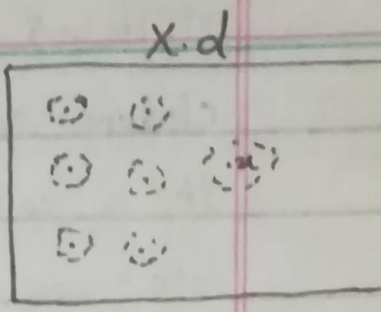


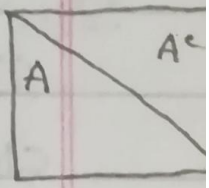
Proof Let $(X.d)$ be a metric space

(i) $\phi \subseteq X$, ϕ is finite, ϕ is closed

$\phi^c = X$ is open

(ii) $X \subseteq X$, X is closed set as $X^d = X$

$X^c = \phi$ is open trivially.



(iii) Let $A \subseteq X$ such that $A \neq \phi$, $A = X$

Suppose 'A' is closed (Given)

A^c is open (to prove)

Let $x \in A^c$ Then $x \notin A$, So, A is closed set $A = A^d$

$x \notin A^d$; 'x' is not a limit point of 'A' Then

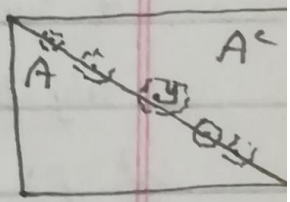
There exist $r > 0$ such That $A \cap S_r(x) = \phi$

$S_r(x) \subseteq A^c$, $\forall\, x \in A^c$, as 'x' is arbitrary point

A^c is open set.

converesly Let A^c is open (Given) to prove A^c is closed

Let A is not closed, $A \neq A^d$



$y \in X$ such That $y \in A^d$ but $y \notin A$

if $y \in A^d$ Then 'y' is limit point

of 'A' Taking a open sphere at 'y' $\forall\, l > 0$

$S_l(y) \cap A \neq \phi$ — I

$y \notin A$ Then $y \in A^c$, A^c is open set, There exist

$p > 0$ such That $S_p(y) \subseteq A^c$, $S_p(y) \notin A$ Then

$S_p(y) \cap A = \phi$ — II So, our contradiction

is wrong.

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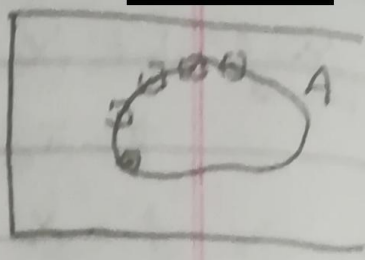
Closure of a set in metric space.

It is collection of all those points which are arbitrary closed to "A"



Closure of $A = \bar{A} = A \cup A^d$

eg $Q^d = R$, $\bar{Q} = Q \cup Q^d = Q \cup R = R$

$(Q^c)^d = R$, $\overline{Q^c} = Q^c \cup (Q^c)^d = Q^c \cup R = R$

Closure of Harmonic Sequence

$A = \left\{\frac{1}{n}\right\}_{n\in N} = \{1, \frac{1}{2}, \frac{1}{3}, \dots\}$, $A^d = \left(\left\{\frac{1}{n}\right\}_{n\in N}\right)^d = \{0\}$

$\bar{A} = A \cup A^d = \{1, \frac{1}{2}, \frac{1}{3}, \dots\} \cup \{0\} = \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\}$

Closure of an open set in R

$A =]0,1[$, $A^d = (0,1)$; $0,1 \notin \overset{\text{open}}{]0,1[}$, $0,1 \in \overset{\text{close}}{[0,1]}$

$\bar{A} = A \cup A^d = (0,1) \cup (0,1)^d = [0,1]$

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lecture - 42

Closeness is a closure Theorem

A subset 'F' of a metric space, is closed if and only if $F = \bar{F}$

Let p: F is a closed set, q: $F = \bar{F} = F \cup F^d$

if 'F' is closed, Then 'F' contain all of its limit point

$F^d \subseteq F$ — I, $F \cup F^d = \bar{F} \cup F$, $\bar{F} \subset F$ — II

By Defination $\bar{F} = F \cup F^d$

$F \subseteq \bar{F}$ — III. from II and III

$\bar{F} \subseteq F \subseteq \bar{F}$, $F = \bar{F}$

$A \subseteq A \cup B$
$A \subseteq B \subseteq A$
$A = B$

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Conversly

Let $F = \bar{F}$ (given) to prove 2; F is closed sets $(\sim 2 \to \sim p)$, $\sim 2 = F$ is not closed set Then There exist a limit points say 'p' of F such That $p \notin F$ Then $p \notin \bar{F}$ because $F = \bar{F}$

$p \notin F \cup F^d$ $\because \bar{F} = F \cup F^d$, $p \notin F^d$ so, p is not a limit point of 'F'

contradiction is wrong so, F is closed

Set Theoretic operation in closures.

Theorem: If 'A' and 'B' are two subset of a metric space (X, d), Then prove That

(i) $A \subseteq B$ Then $\bar{A} \subseteq \bar{B}$ (ii) $\overline{(A \cup B)} = \bar{A} \cup \bar{B}$

(iii) $\overline{(A \cap B)} = \bar{A} \cap \bar{B}$ (iv) $\bar{A}$ is largest closed super set of A

Proof (i) given $A \subseteq B$ Then $A^d \subseteq B^d$ taking a union $A \cup A^d \subseteq B \cup B^d$ Then $\bar{A} \subseteq \bar{B}$

(ii) L.H.S $= \overline{A \cup B}$

$= (A \cup B) \cup (A \cup B)^d$ $\qquad \bar{A} = A \cup A^d$

$= (A \cup B) \cup (A^d \cup B^d)$

$= (A \cup A^d) \cup (B \cup B^d)$

$= \bar{A} \cup \bar{B}$

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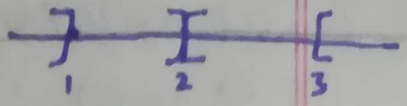
(iii) $\overline{(A \cap B)} = \bar{A} \cap \bar{B}$, $A \cap B \subseteq A$ and $A \cap B \subseteq B$

$\overline{A \cap B} \subseteq \bar{A}$ and $\overline{A \cap B} \subseteq \bar{B}$

taking intersection: $\overline{(A \cap B)} \cap \overline{(A \cap B)} \subseteq \bar{A} \cap \bar{B}$

$(\overline{A \cap B}) \subseteq \bar{A} \cap \bar{B}$

e.g Let $A=(1,2)$, $B=(2,3)$



$\bar{A}=[1,2]$ $\bar{B}=[2,3]$

$L.H.S = (\overline{A \cap B}) = \overline{(1,2) \cap (2,3)} = \bar{\phi} = \phi \quad \because F=\bar{F}$

$R.H.S = \bar{A} \cap \bar{B} = [1,2] \cap [2,3] = \{2\}$

So, $\phi \subseteq \{2\}$

$(\overline{A \cap B}) \subseteq \bar{A} \cap \bar{B}$

II For any $\phi \neq A \subseteq X$, $\bar{A}$ is closed

$\bar{A} = A \cup A^d$, $A \subseteq \bar{A}^d$ — I

Let $\phi \neq B \subseteq X$, be arbitrary any other closed set containing 'A' , $A \subseteq B$, $\bar{A} \subseteq \bar{B}$, $\bar{A} \subseteq B$

from I and II $A \subseteq \bar{A} \subseteq B$ $\because \bar{B}=B$ (2)

$\bar{A}$ is smallest closed super set of A

Let $\phi \neq B \subseteq X$ be any other arbitrary closed set contain in 'A' $B \subseteq A$ then $\bar{B} \subseteq \bar{A}$, $B \subseteq \bar{A}$ — III

from I, III $B \subseteq A \subseteq \bar{A}$, $\bar{A}$ is largest closed super set of A

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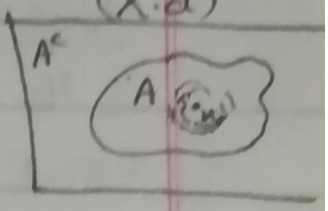
Interior, exterior and Boundary of a set in metric space (X.d)



Interior $x \in S_r(x) \subseteq A$

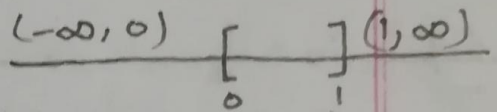
$Int(A) = \{x \in A, A \subseteq X, S_r(x) \subseteq A \text{ for some } r > 0\}$

exterior $Ext(A) = \{x \in A^c, A^c \subseteq X, S_r(x) \subseteq A^c \text{ for some } r>0\}$

$ext(A) = Int(A^c)$

Boundary Frontier $Fr(A) = \{x \in X ; Int(A) \cap S_r(x) \neq \phi, ext(A) \cap S_r(x) \neq \phi, \forall r > 0\}$

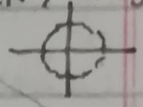
e.g $A = [0,1]$



$Int(A) = (0,1)$, $ext(A) = (-\infty, 0) \cup (1, \infty)$

$Fr(A) = \{0,1\}$

$A = \{(x,y) \in R^2, x^2+y^2 = 1\}$



e.g $Int\{1\} \neq \phi$, $Int\, \mathbb{Z} = \phi$

e.g Discrete metric space (X.d)

Say $A \subseteq X$, $Int(A) = \bigcup_{x \in A} \{x\} = A$

$ext(A) = \bigcup_{x \in A^c} \{x\} = A^c$

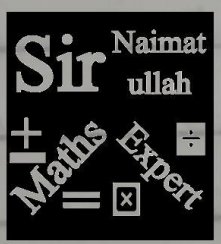
$Fr(A) = \phi$

$Int(A), x^2+y^2 < 1$

$ext(A), x^2+y^2 > 1$

$Fr(A), x^2+y^2 = 1$

Theorem $Int(A) \cup ext(A) \cup Fr(A) = X$



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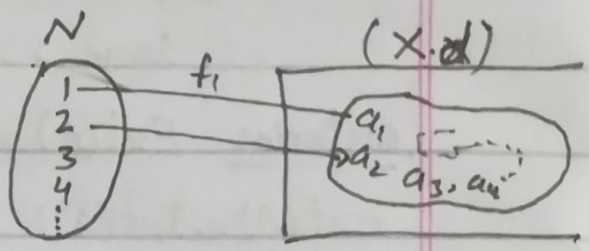
Sequences in a metric space.

Sequence infinit list of numbers.

"It is a function whose domain is set of Natural numbers and whose range is subset of a metric space

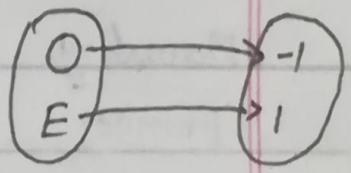
$\{a_n\}_{n=1}^{\infty}$, $(a_n) = \{a_1, a_2, \dots\}$

$\{a_n\}$, $a_n \rightarrow$ nth term



e.g $f(n) = (-1)^n$ $n \in N$

$f(1) = -1$, $f(2) = 1$ .....



e.g $a_n = \frac{1}{n}$, $a_n = n^2$

Montonicity of sequences.

$a_n = \frac{1}{n}$ ↑

(i) increasing sequences $a_{n+1} \geq a_n \quad \forall\ n \in N$

ii) Decreasing sequences $a_{n+1} \leq a_n \quad \forall\ n \in N$

$a_n = n^2$ ↓

Constant sequences.

It is not increasing and decreasing e.g $a_n = 3$

$a_n = a_\infty$ = No such term exist $\infty \notin N$ $n \in N$

Bounded sequences let $(X.d)$ be a metric space an $\{x_n\}$ be a sequence in X, Then its is said to be bounded if There exist $m \in \mathbb{R}$ $m < \infty$ such That $d(x_m, x_n) < M \quad \forall\ m, n \in N$

e.g $d: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R} : d(x,y) = |x-y|, \ \forall\ x, y \in \mathbb{R}$

$\{x_n = \frac{1}{n}\}_{n \in N}^{\infty} : d(x_m, x_n) = \left|\frac{1}{m} - \frac{1}{n}\right| < 1$

it is bounded sequence

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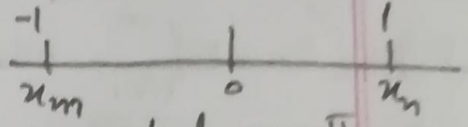
e.g $\{a_n = (-1)^n\}_{n=1}^{\infty}$, $n \in N$ is bounded sequences

because $-1 \leq a_n \leq 1 \quad \forall\, n \in N$

-1 is Lower bound and 1 is upper bound.

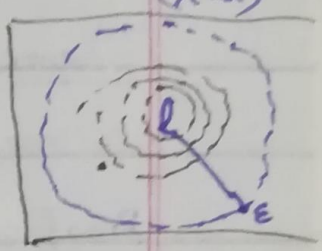
$d(x_m, x_n) \leq 2$



e.g $\{a_n = n\}_{n=1}^{\infty}$ $n \in N$ it is unbounded as There not exist any $K \in R$ such That $|a_n - a_m| < K$

Convergent Sequence in a metric space.

A sequence $\{x_n\}$ in a metric space $(X.d)$ is said to be convergence to limit 'l' if $\forall$ open spheres at 'l' There exist $n_0 \in N$ such That $x_n \in S_\varepsilon(l)$



$\forall\, n \geq n_0$

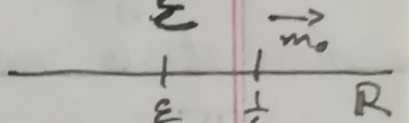
Symbolically $\{x_n\} \to l$, $x_n \to l$ $\lim_{n\to\infty} x_n = l$

e.g $\{a_n = \frac{1}{n}\}$ Show That $\lim_{n\to\infty} \frac{1}{n} = 0$

$\forall\, \varepsilon > 0$ taking $|a_n - 0| < \varepsilon$, $|\frac{1}{n} - 0| < \varepsilon$, $\frac{1}{n} < \varepsilon$

$n > 0$, $|\frac{1}{n}| = \frac{1}{n}$, so, $\frac{1}{n} < \varepsilon$, $n > \frac{1}{\varepsilon}$

if m_0 is such a natural n_0



Just $\geq \frac{1}{\varepsilon}$, Then $|a_n - 0| < \varepsilon$, $\forall\, n \geq m_0$ Just $\geq \frac{1}{\varepsilon}$

$a_n \to 0$

e.g $\{\frac{1}{5^n}\} \to 0$ let the given $\varepsilon > 0$, taking $|a_n - 0| < \varepsilon$

$|\frac{1}{5^n} - 0| < \varepsilon$, $\frac{1}{5^n} < \varepsilon$, $5^n < \frac{1}{\varepsilon}$

$\log_5 5^n \leq \log_5(\frac{1}{\varepsilon})$, $n \log_5 5 < \log_5(\frac{1}{\varepsilon})$

$n < \log_5(\frac{1}{\varepsilon})$; if "$m_0$" is a natural n_0

Just $\geq \log_5(\frac{1}{\varepsilon})$ Then $\{\frac{1}{5^n}\} \to 0$

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Divergent Sequence A sequence $\{x_n\}$ in a metric space $(X.d)$. However large $l>0$, There exist $n_0 \in N$ such That $n \geq n_0$. $\forall\ n \in N, a_n \notin S_x(l)$

e.g $\{a_n = n\}_{n=1}^{\infty}$ $n \in N$ $a_n \to \infty$

e.g $\{a_n = 5^n\}_{n=1}^{\infty}$ $n \in N$ $a_n \to \infty$

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Uniqueness of a limit of a sequence in a metrice Space.

Theorem limit of a convergent sequence in a metric space is unique.

Proof Let $\{x_n\} \to x$ and $\{x_{kn}\} \to y$ in a metric space $(X.d)$ such That $x \neq y \Rightarrow d(x,y) \neq 0$

For any $\varepsilon > 0$ There exist $x \in N$ such That

$\frac{d(x,y)}{n} < \varepsilon$

$\frac{d(x,y)}{n} = 0 < \varepsilon$

$t < \varepsilon$, $t = 0$

it is not possible so, $x = y$

Theorem Every convergent sequence in a metric space is bounded but converse may not true.

Proof in $(X.d)$ let $\{x_n\}$ be a ~~metric space~~ a convergente sequence such That $x_n \to x \in N$

$\forall\ \varepsilon > 0$, there exist $n_0 \in N$ such That $\forall n \geq n_0$

$d(x_n, x_0) < \varepsilon$ —①

Let $K = \max\{d(x_1, x) + \varepsilon, d(x_2, x) + \varepsilon, \ldots d(x_{n-1}, x)\}$

$d(x_n, x) < k$, $\forall n \in N$, $\{x_n\}_{n=1}^{\infty} \subseteq S_k(x)$

$d(x_m, x_n) < k$, $\forall m, n \in N$

So, $\{x_n\}_{n=1}^{\infty}$ is bounded sequence

conversly in (R, d)

$\{x_n\} = \{(-1)^n\}_{n=1}^{\infty}$ is a bounded sequence

Here $-1 \leq a_n \leq 1$ $\forall n \in N$

$a_n \to 1$ if n is even, $a_n \to -1$ if n is odd

limit of the sequence is not unique and

$\{x_n\}$ is not convergence

convergence $\Rightarrow$ bounded

Bounded $\not\Rightarrow$ convergence.

Theorem In a metric space (X, d), if $\{x_n\} \to x$

$\{y_n\} \to y$ Then $d(x_n, y_n) \to d(x, y)$

Proof $\{x_n\} \to x$ $\forall \varepsilon > 0$, There exist $n_1 \in N$

such That $\forall n \geq n_1$, $d(x_n, x) < \varepsilon$ —— I

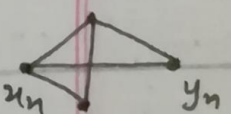
$\{y_n\} \to y$ $\forall \varepsilon > 0$, There exis $n_2 \in N$ such That

$\forall n \geq n_2$ $d(y_n, y) < \varepsilon$ —— (2)

Let $n_0 = \max\{n_1, n_2\}$

from (1), and (2) $d(x_n, x) < \frac{\varepsilon}{2}$, $d(y_n, y) < \frac{\varepsilon}{2}$ $\forall n \geq n_0$

$|d(x_n, y_n) - d(x, y)| \leq |d(x_n, x) + d(x, y) + d(y, y_n) - d(x, y)|$



$\leq |d(x_n, x) + d(y_n, y)|$

$$\leq d(x_n, x) + d(y_n, y)$$

$$\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2}, \quad \leq \frac{2\varepsilon}{2} = \varepsilon$$

$$|d(x_n, y_n) - d(x, y)| < \varepsilon \quad \forall\ n \geq n_0$$

$$d(x_n, y_n) \longrightarrow d(x, y)$$

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